

数列

例1: 已知数列 $\{a_n\}$ 的递推关系且 $a_1=2$, 请求出 $\{a_n\}$ 通项公式

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| (1) $a_{n+1} = a_n + 2$ | (2) $a_{n+1} = a_n + 2n$ | (3) $a_{n+1} = a_n + 2^n$ |
| (4) $a_{n+1} = a_n + \ln(1 + \frac{1}{n})$ | (5) $a_{n+1} = 2a_n + 2$ | (6) $a_{n+1} = 2a_n + 2n$ |
| (7) $a_{n+1} = 2a_n + 2^n$ | (8) $a_{n+1} = 3a_n + 2^n$ | (9) $a_{n+1} = \frac{n}{n+2} a_n$ |
| (10) $a_{n+1} = \frac{2n+1}{2n-1} a_n$ | (11) $a_{n+1} + a_n = 2n$ | (12) $a_{n+1} + a_n = 2^n$ |
| (13) $a_{n+1} + a_n = n \cdot 2^n$ | (14) $a_{n+1} = \frac{a_n}{a_n+1}$ | (15) $a_{n+1} = \frac{a_n}{a_n+2}$ |
| (16) $a_{n+1} = \frac{4a_n-1}{2a_n+1}$ | (17) $a_{n+1} = \frac{a_n-1}{a_n+3}$ | (18) $a_{n+1} = a_n^2$ |
| (19) $a_{n+2} = 5a_{n+1} - 6a_n, a_2=3$ | (20) $a_{n+2} = 4a_{n+1} - 4a_n, a_2=3$ | (21) 当 n 为奇数时, $a_{n+1} = 2a_n$;
当 n 为偶数时, $a_{n+1} = a_n + 1$ |
| (22) $a_{n+1} + (-1)^n a_n = 2n$ | (23) $a_{n+1} \cdot a_n = 4n^2 - 1$ | |

例2: 已知数列 $\{a_n\}$ 通项公式, 请求出 $\{a_n\}$ 前 n 项和 S_n

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| (1) $a_n = \frac{1}{n(n+1)}$ | (2) $a_n = \frac{1}{n(n+2)}$ | (3) $a_n = \frac{1}{4n^2-1}$ |
| (4) $a_n = \frac{n^2}{4n^2-1}$ | (5) $a_n = \frac{1}{\sqrt{n+1} + \sqrt{n}}$ | (6) $a_n = \frac{1}{(n+1)\sqrt{n} + n\sqrt{n+1}}$ |
| (7) $a_n = \sqrt{1 + \frac{1}{n^2} + \frac{1}{(n+1)^2}}$ | (8) $a_n = \frac{1}{n(n+1)(n+2)}$ | (9) $a_n = \frac{2 \times 3^n}{(3^n+1)(3^{n+1}+1)}$ |
| (10) $a_n = \frac{n+2}{n(n+1) \cdot 2^n}$ | (11) $a_n = \frac{n \cdot 2^n}{(n+1)(n+2)}$ | (12) $a_n = \frac{2n+1}{n^2 \cdot (n+1)^2}$ |
| (13) $a_n = (-1)^n (4n-3)$ | (14) $a_n = (-1)^n (4n-3)^2$ | (15) $a_n = n \cdot 2^n$ |
| (16) $a_n = \frac{n}{2^n}$ | (17) $a_n = n \cos(\frac{n\pi}{2}) + 1$ | (18) $a_n = 23-7n $ |
| (19) 当 n 为奇数时, $a_n = 2n$; 当 n 为偶数时, $a_n = 2^n$ | | |

例3: 已知正项数列 $\{a_n\}$, 记 $\{a_n\}$ 前 n 项和为 S_n , 求出下列对应条件 a_n

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| (1) $4S_n = (a_n + 1)^2$ | (2) $4S_n = (a_{n+1} - 1)^2, a_1=1$ | (3) $4S_{n+1} = (a_n + 3)^2, a_1=1$ |
| (4) $S_n = \frac{a_n}{2} + \frac{1}{a_n} - 1$ | (5) $2S_n + 1 = 2a_n^2 + a_n$ | (6) $\frac{S_{n+1}}{n+1} - \frac{S_n}{n} = 1, a_1=1$ |
| (7) $a_n = \frac{S_n}{n} + n - 1, a_1=1$ | (8) $\sqrt{S_{n+1}} - \sqrt{S_n} = 1, a_1=1$ | (9) $a_{n+1}(S_n + S_{n+1}) = 1, a_1=1$ |
| (10) $\frac{S_n}{a_{n+1}} - \frac{S_n}{a_n} = -\frac{1}{2}, a_1=1$ | (11) $S_n + S_{n+1} = (n+1)^2, a_1=1$ | (12) $\sum_{i=1}^n a_i^3 = S_n^2$ |
| (13) $a_{n+1} a_n = 4S_n, a_1=1$ | (14) $a_{n+1} + S_n S_{n+1} = 0, a_1=1$ | (15) $\sum_{i=1}^n \frac{1}{a_i \cdot a_{i+1}} = \frac{1}{6} - \frac{1}{3a_{n+1}}, a_1=a_2$ |
| (16) $(1+\frac{1}{a_1}) \cdot (1+\frac{1}{a_2}) \cdots (1+\frac{1}{a_n}) = 2^{\frac{n(n+1)}{2}}$ | | |

例4: 已知数列 $\{a_n\}$ 是等比数列, 数列 $\{b_n\}$ 满足下列式子, 请求出 $\{a_n\}$ 、 $\{b_n\}$ 通项公式

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| (1) $a_1=2, b_2=5$, 且对于任意 $n \in \mathbb{N}^*$ 都有: $\sum_{i=1}^n a_i \cdot b_i = 2a_n(b_n - 3) + 8$ |
| (2) $a_1=\frac{1}{2}, b_2=5$, 且对于任意 $n \in \mathbb{N}^*$ 都有: $\sum_{i=1}^n a_i \cdot b_{n+1-i} = 4a_n + b_n - 3$ |

例5: 已知数列 $\{a_n\}$ 、 $\{b_n\}$ 满足: $a_n + b_{n+1} = 2^n$, $a_{n+1} + b_n = 2n$, 且 $a_1=b_1=0$, 求数列 $\{a_n\}$ 、 $\{b_n\}$ 通项公式

例6: 已知 S_n 是公差为0的等差数列 $\{a_n\}$ 前 n 项和, 数列 $\{b_n\}$ 满足下列式子, 求出 $\{a_n\}$ 、 $\{b_n\}$ 通项公式

(1) $a_1=3, b_2=2$, 且对于 $\forall n \in \mathbb{N}^*$ 都有: $\frac{a_n - b_n}{n} = 2$

(2) $a_3=4, a_4=S_3$, 且对于 $\forall n \in \mathbb{N}^*$ 都有: $S_n + b_n, S_{n+1} + b_n, S_{n+2} + b_n$ 成等比数列

(3) $a_1=1$, $\{b_n\}$ 是等差数列, 且对于 $\forall n \in \mathbb{N}^*$ 都有: $b_n = \frac{S_n}{a_n}$

例7: 已知数列 $\{a_n\}$ 前 n 项和 S_n , $\{S_n\}$ 的前 n 项积为 T_n , 请求出 $\{a_n\}$ 通项公式

(1) $\frac{2}{S_n} + \frac{1}{T_n} = 2$

(2) $\frac{2}{S_n} + \frac{1}{T_n} = 1$

(3) $T_n = A_{2n}^n \cdot S_n (n \geq 2)$

(4) $T_n = 2^{n(n+1)} \cdot S_n (n \geq 2)$

(5) $T_n = C_{2n}^n \cdot S_n (n \geq 2)$

例8: 已知正项数列 $\{a_n\}$, 从下列式子中任选两个作为条件, 证明另一个式子成立

① $\{a_n\}$ 是等差数列

② $\{\sqrt{S_n}\}$ 是等差数列

③ $a_2 = 3a_1$

例9: 已知数列 $\{a_n\}$ 前 n 项和为 S_n , 证明下式成立

(1) $a_n = \frac{1}{n^2}$, 证: $S_n < \frac{7}{4}$

(2) $a_n = \frac{1}{\sqrt{n}}$, 证: $2(\sqrt{n+1}-1) < S_n < 2\sqrt{n}$

(3) $a_n = \frac{1}{n\sqrt{n}}$, 证: $S_n < 3$

(4) $a_n = \frac{2^n}{(2^n-1)^2}$, 证: $S_n < 3$

(5) $a_n = \frac{2}{3^n-1}$, 证: $S_n < \frac{3}{2}$

(6) $a_n = \frac{6^n}{(n+1)(2n+1)}$, 证: $S_n \leq \frac{6^n-1}{5}$

(7) $a_n = \ln(1+\frac{1}{n^2})$, 证: $S_n < 2$

(8) $a_n = \frac{\ln(n)}{n+1}$, 证: $S_n \leq \frac{n(n-1)}{4}$

(9) $a_n = \frac{1}{\sqrt{n^2+n}}$, 证: $S_n > \ln(n+1)$

(10) $a_n = \ln(1+\frac{1}{n})$, 证: $\sum_{i=1}^n \frac{1}{i+1} < S_n < \sum_{i=1}^n \frac{1}{i}$

例10: 证明下列命题成立

(1) 已知 $\{a_n\}$ 是公比为2的等比数列, 求证: $\{a_n\}$ 中任意三项均不构成等差数列

(2) 已知 $\{a_n\}$ 是公比为2的等比数列, 求证: $\{a_n+1\}$ 不是等比数列

(3) 已知正实数 a, b, c 构成公差为0的等差数列, 求证: $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ 不是等差数列

(4) 已知正实数 a, b, c 构成公差为0的等差数列, 求证: $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ 不是等比数列

例11: 完成下列问题

(1) 已知 $a_n = 3n-1, b_n = 2n+1$, 将 $\{a_n\}$ 与 $\{b_n\}$ 的公共项记为 $\{c_n\}$, 求 $\{c_n\}$ 通项公式

(2) 已知 $a_n = 3n-1, b_n = 2^n$, 将 $\{a_n\}$ 与 $\{b_n\}$ 的公共项记为 $\{c_n\}$, 求 $\{c_n\}$ 通项公式

(3) 已知 $a_n = 2n-1, b_n = 2^n-1$, 将 $\{a_n\}$ 与 $\{b_n\}$ 的公共项去掉, 剩下的项从小到大依次排列成数列 $\{c_n\}$, 求 $\{c_n\}$ 前20项和 S_{20}

(4) 已知 $a_n = 2n-1, b_n = 2^n$, 在 a_k 与 a_{k+1} 之间插入 k 个 $b_k (k \in \mathbb{N}^*)$ 组成新数列 $\{c_n\}$, 求 $\{c_n\}$ 前25项和 S_{25}

(5) 已知 $a_n = 2n+1, b_n = 2^n$, 将 $\{a_n\}$ 与 $\{b_n\}$ 所有项从小到大依次排列成数列 $\{c_n\}$, 求 $\{c_n\}$ 前27项和 S_{27}

(6) 已知 $a_n = 3n - 1$, $b_n = 2^n$, 将 $\{a_n\}$ 的项落在区间 $[b_k, b_{k+1}]$ 上的个数记为 c_k , 求 $\{c_k\}$ 的通项公式

(7) 已知 $a_n = 2^n$, 在 a_k 与 a_{k+1} 之间插入 k 个数 ($k \in \mathbb{N}^*$) 使得这 $k+2$ 个数组成等差数列, 记该等差数列公差为 b_k , 求 $\{b_k\}$ 的通项公式

例 12: 已知 $[x]$ 表示不超过 x 的最大整数, 如 $[2.1] = 2$, 若数列 $\{a_n\}$ 满足: $a_n = [\frac{2^n}{3}]$ ($n \in \mathbb{N}^*$), 记 $\{a_n\}$ 前 n 项和为 S_n , 求 S_n 通项公式

例 13: 已知数列 $\{a_n\}$ 满足: $a_1 = \frac{1}{2}$, $a_{n+1} = \frac{1}{1+a_n}$ ($n \in \mathbb{N}^*$)

(1) 判断数列 $\{a_{2n}\}$ ($n \in \mathbb{N}^*$) 的单调性

(2) 求证: $\frac{1}{2} \leq a_n < 1$

(3) 求证: $|a_{n+1} - a_n| \leq \frac{1}{6} \times (\frac{2}{5})^{n-1}$

例 14: 已知数列 $\{a_n\}$ 首项不为 0, 记 $\{a_n\}$ 前 n 项和为 S_n , 若对于 $\forall n \in \mathbb{N}^*$ 都有: $2\sqrt{S_n} = a_n + c$ (c 为常数)

(1) 求证: $c \leq 1$

(2) 若 $c = 0$, 求证: $S_n \geq (n+1)^2$

(3) 若 $\{a_n\}$ 为等比数列, 求 c 值

例 15: 已知各项均大于 1 的数列 $\{a_n\}$ 满足: $a_1 = 2$, 对于 $\forall n \in \mathbb{N}^*$ 都有: $a_{n+1} = a_n - a_n^2 + 1$, 记数列 $\{\frac{1}{a_n}\}$ 的前 n 项和与 $\{a_n\}$ 前 n 项和分别为 S_n 和 T_n

(1) 求证: $\{\frac{1}{a_n}\}$ 是递减数列

(2) 求证: $T_n \leq \frac{1}{2} \leq S_n$

(3) 求证: $\frac{1}{T_n} - S_n > 1$

例 16: 已知数列 $\{a_n\}$ 满足: $a_1 = \frac{1}{2}$, 对于 $\forall n \in \mathbb{N}^*$ 都有: $a_{n+1} = a_n - a_n^2$

(1) 求证: $1 < \frac{1}{a_{n+1}} - \frac{1}{a_n} \leq 2$

(2) 求证: $\sum_{i=1}^n a_i^2 \leq \frac{n}{2(n+1)}$

例 17: 已知数列 $\{a_n\}$ 满足: $a_1 = m$ ($m \in \mathbb{N}^*$), 对于 $\forall n \in \mathbb{N}^*$ 都有:

当 a_n 为奇数时, $a_{n+1} = 3a_n + 1$; 当 a_n 为偶数时, $a_{n+1} = \frac{a_n}{2}$

(1) 当 $m = 17$ 时, 求: 使 $a_n = 1$ 时的最小正整数 n

(2) 若 $a_8 = 1$, 求 m 所有可能的取值范围的集合

例 18: 已知数列各项均为正整数的 $\{a_n\}$ 对于 $\forall n \in \mathbb{N}^*$ 都有:

当 a_n 为奇数时, $a_{n+1} = a_n + 3$; 当 a_n 为偶数时, $a_{n+1} = \frac{a_n}{2}$

(1) 若 $a_4 = 1$, 求 a_1 所有可能取值范围的集合

(2) 若 $\exists m \in \mathbb{N}^*$, 使得 $a_m = 3k (k \in \mathbb{N}^*)$, 求证: 对于 $\forall n \in \mathbb{N}^*$ 都有 $\frac{a_n}{3}$ 为整数

例 19: 在正项递增数列 $\{a_n\}$ 中, $a_1 = 1$, 对于 $\forall k \in \mathbb{N}^*$ 都有: $a_{2k-1}, a_{2k}, a_{2k+1}$ 成公差为 $2k$ 的等差数列

(1) 求 $\{a_n\}$ 的通项公式

(2) 是否存在实数 m , 使得对于 $\forall k \in \mathbb{N}^*$ 都有: $a_{2k} + m, a_{2k+1} + m, a_{2k+2} + m$ 成等比数列, 若存在, 请求出 m 值; 若不存在, 请说明理由

例 20: 在正项递增数列 $\{a_n\}$ 中, $a_1 = 4$, 对于 $\forall k \in \mathbb{N}^*$ 都有: $a_{2k-1}, a_{2k}, a_{2k+1}$ 成公比为 $1 + \frac{1}{k}$ 的等比数列

(1) 求 $\{a_n\}$ 的通项公式

(2) 记 $b_n = (-1)^n a_{2n-1} (n \in \mathbb{N}^*)$ 数列 $\{b_n\}$ 的前 n 项和为 S_n , 求 S_n 的通项公式

例 21: 已知无穷数列 $\{a_n\}$ 满足: a_1, a_2 均为正整数, 对于 $\forall n \in \mathbb{N}^*$ 都有: $a_n = |a_{n+1} - a_{n+2}|$

(1) 若 $a_1 = 1, a_3 = 2$, 求 a_4 的值

(2) 求证: "存在 $k \in \mathbb{N}^*$, 使得 $a_k = 0$ " 是 " $\{a_n\}$ 是周期为 3 的数列" 的必要不充分条件

例 22: 已知数列 $a_1, a_2, \dots, a_{4m+2} (m \in \mathbb{N}^*)$ 是公差为 $d \neq 0$ 的等差数列, 若从中删去两项 a_i 和 $a_j (i < j, i, j \in \mathbb{N}^*)$ 后, 剩余的 $4m$ 项可被平均分为 m 组, 且每组的 4 个数都能构成等差数列, 则称数列 $a_1, a_2, \dots, a_{4m+2}$ 为 (i, j) -可分数列

(1) 写出所有的 $(i, j) (1 \leq i < j \leq 6)$, 使得 $a_1, a_2, \dots, a_6$ 为 (i, j) -可分数列

(2) 当 $m \geq 3$ 时, 求证: $a_1, a_2, \dots, a_{4m+2}$ 是 $(2, 13)$ -可分数列

例 23: 定义 R_p 数列 $\{a_n\}$ 对于实数 p 满足下列性质:

① $a_1 + p \geq 0, a_2 + p = 0$; ② $\forall n \in \mathbb{N}^*$ 都有: $a_{4n-1} \leq a_{4n}$; ③ $\forall m, n \in \mathbb{N}^*$ 都有: $a_{m+n} \in \{a_m + a_n + p, a_m + a_n + p + 1\}$

(1) 判断前四项分别为: $2, -2, 0, 1$ 的数列 $\{a_n\}$ 是否为 R_2 数列, 并说明理由

(2) 若 $\{a_n\}$ 是 R_0 数列, 求 a_5 值

例 24: 记数列 $\{a_n\}$ 的前 n 项和为 S_n , 若对于 $\forall n \in \mathbb{N}^*$, 有且仅有一个 $m (m \in \mathbb{N}^*)$, 使得 $S_m \leq a_n < S_{m+1}$, 则称 $\{a_n\}$ 为 "X 数列", 记 $b_n = S_{n+1} - a_n (n \in \mathbb{N}^*)$, 称 $\{b_n\}$ 为 $\{a_n\}$ 的 "余项数列"

(1) 若 $\{a_n\}$ 的前四项分别为: $0, 1, -1, 1$, 判断 $\{a_n\}$ 是否为 "X 数列", 并说明理由

(2) 若 $S_n = 2^n$, 求证: $\{a_n\}$ 为 "X 数列", 并求出 $\{a_n\}$ 的 "余项数列" $\{b_n\}$ 通项公式

数列.

例1: (1) 由 $a_{n+1} - a_n = 2$, 则 $\begin{cases} a_{n+1} - a_n = 2 \\ a_2 - a_1 = 2 \end{cases}$ 则 $a_{n+1} - a_1 = 2 \times n$ 则 $a_{n+1} = 2n + 2 = 2(n+1)$

当 $n \geq 2$ 时, $a_n = 2n$; 当 $n=1$ 时, $a_1 = 2$ 符合左式 即 $a_n = 2n$

(2) 由 $a_{n+1} - a_n = 2n$, 则 $\begin{cases} a_{n+1} - a_n = 2n \\ a_2 - a_1 = 2 \times 1 \end{cases}$ 则 $a_{n+1} - a_1 = 2(1+2+\dots+n) = 2 \times \frac{(1+n)n}{2} = n(n+1)$

则 $a_{n+1} = n(n+1) + 2$ 当 $n \geq 2$ 时, $a_n = n(n-1) + 2$; 当 $n=1$ 时, $a_1 = 2$ 符合左式, 即 $a_n = n^2 - n + 2$

(3) 由 $a_{n+1} - a_n = 2^n$, 则 $\begin{cases} a_{n+1} - a_n = 2^n \\ a_2 - a_1 = 2^1 \end{cases}$ 则 $a_{n+1} - a_1 = 2^1 + 2^2 + \dots + 2^n = \frac{2(2^n - 1)}{2 - 1} = 2^{n+1} - 2$

则 $a_{n+1} = 2^{n+1}$ 当 $n \geq 2$ 时, $a_n = 2^n$; 当 $n=1$ 时, $a_1 = 2$ 符合左式, 即 $a_n = 2^n$

(4) 由 $a_{n+1} - a_n = \ln(1 + \frac{1}{n}) = \ln(\frac{n+1}{n}) = \ln(n+1) - \ln(n)$ 则 $\begin{cases} a_{n+1} - a_n = \ln(n+1) - \ln(n) \\ a_2 - a_1 = \ln 2 - \ln 1 \end{cases}$ 则 $a_{n+1} - a_1 = \ln(n+1)$

则 $a_{n+1} = \ln(n+1) + 2$ 当 $n \geq 2$ 时, $a_n = \ln(n) + 2$; 当 $n=1$ 时, $a_1 = 2$ 符合左式, 即 $a_n = \ln(n) + 2$

(5) 由 $a_{n+1} = 2a_n + 2$, 设 $a_{n+1} + \lambda = 2(a_n + \lambda)$ 则 $a_{n+1} = 2a_n + \lambda$ 则 $\lambda = 2$ 即 $a_{n+1} + 2 = 2^{n+1}(a_1 + 2)$

则 $a_n = 2^{n+1} - 2$

(6) 由 $a_{n+1} = 2a_n + 2n$, 设 $a_{n+1} + \lambda(n+1) + \mu = 2[a_n + \lambda n + \mu]$, 则 $a_{n+1} = 2a_n + \lambda n + \mu - \lambda$ 则 $\begin{cases} \lambda = 2 \\ \mu - \lambda = 0 \end{cases}$

即 $\begin{cases} \lambda = 2 \\ \mu = 2 \end{cases}$ 则 $a_{n+1} + 2(n+1) + 2 = 2^{n+1}(a_1 + 2 \times 1 + 2)$ 则 $a_n = 3 \times 2^n - 2n - 2$

(7) 由 $a_{n+1} = 2a_n + 2^n$, 同除 2^{n+1} 则 $\frac{a_{n+1}}{2^{n+1}} = \frac{a_n}{2^n} + \frac{1}{2}$ 则 $\{\frac{a_n}{2^n}\}$ 是首项为 $\frac{a_1}{2} = 1$, 公差为 $\frac{1}{2}$ 的等差数列.

则 $\frac{a_n}{2^n} = \frac{1}{2}(n-1) + 1 = \frac{n+1}{2}$ 即 $a_n = (n+1)2^{n-1}$

(8) 由 $a_{n+1} = 3a_n + 2^n$ 设 $a_{n+1} + \lambda \cdot 2^{n+1} = 3(a_n + \lambda \cdot 2^n)$ 则 $a_{n+1} = 3a_n + 3\lambda \cdot 2^n - \lambda \cdot 2^{n+1} = 3a_n + \lambda \cdot 2^n$

则 $\lambda = 1$, 则 $a_{n+1} + 2^{n+1} = 3^{n+1}(a_1 + 2^1)$ 则 $a_n = 4 \times 3^{n-1} - 2^n$

(9) 由 $\frac{a_{n+1}}{a_n} = \frac{n}{n+2}$, 当 $n \geq 2$ 时, $\frac{a_{n+1}}{a_n} \cdot \frac{a_n}{a_{n-1}} \cdot \dots \cdot \frac{a_2}{a_1} = \frac{n}{(n+2)} \cdot \frac{(n-1)}{n+1} \cdot \frac{(n-2)}{n} \cdot \frac{(n-3)}{n-1} \cdot \dots \cdot \frac{2}{4} \cdot \frac{1}{3} = \frac{2}{(n+1) \cdot 2(n+2)}$

则 $\frac{a_{n+1}}{a_1} = \frac{2}{(n+1)(n+2)}$ 则 $a_{n+1} = \frac{4}{(n+1)(n+2)}$, 当 $n \geq 2$ 时, $a_n = \frac{4}{n(n+1)}$; 当 $n=1$ 时, $\frac{a_2}{a_1} = \frac{1}{3}$ 则 $a_2 = \frac{2}{3}$

则 a_1, a_2 符合 a_n 式子. 则 $a_n = \frac{4}{n(n+1)}$

(10) 由 $\frac{a_{n+1}}{a_n} = \frac{2n+1}{2n-1}$ 则 $\frac{a_{n+1}}{a_n} \cdot \frac{a_n}{a_{n-1}} \cdot \frac{a_{n-1}}{a_{n-2}} \cdot \frac{a_{n-2}}{a_{n-3}} = \frac{a_{n+1}}{a_1} = \frac{2n+1}{2n-1} \cdot \frac{2n-1}{2n-3} \cdot \frac{2n-3}{2n-5} \cdot \frac{2n-5}{2n-7} = 2n+1$ 则 $a_{n+1} = 2(2n+1)$

当 $n \geq 2$ 时, $a_n = 2(2n-1) = 4n-2$ 当 $n=1$ 时, $a_1 = 2$ 符合左式 则 $a_n = 4n-2$

(11) 由 $a_{n+1} = -a_n + 2n$ 设 $a_{n+1} + \lambda(n+1) + \mu = -(a_n + \lambda n + \mu)$ 则 $a_{n+1} = -a_n - 2\lambda n - 2\mu - \lambda$

则 $\begin{cases} -2\lambda = 2 \\ -2\mu - \lambda = 0 \end{cases}$ 则 $\begin{cases} \lambda = -1 \\ \mu = \frac{1}{2} \end{cases}$ 则 $a_{n+1} - n + \frac{1}{2} = (-1)^{n+1}(a_1 - 1 + \frac{1}{2})$ 则 $a_n = \frac{3}{2} \times (-1)^{n-1} + n - \frac{1}{2}$

(12) 由 $a_{n+1} = -a_n + 2^n$ 设 $a_{n+1} + \lambda \cdot 2^{n+1} = -(a_n + \lambda \cdot 2^n)$ 则 $a_{n+1} = -a_n - 3\lambda \cdot 2^n$

则 $-3\lambda = 1$ 则 $\lambda = -\frac{1}{3}$ 则 $a_{n+1} - \frac{1}{3} \times 2^{n+1} = (-1)^{n+1}(a_1 - \frac{1}{3} \times 2^1)$ 则 $a_n = \frac{4}{3} \times (-1)^{n-1} + \frac{1}{3} \times 2^n$

(13) 由 $a_{n+1} = -a_n + n \cdot 2^n$ 设 $a_{n+1} + [\lambda(n+1) + \mu] \times 2^{n+1} = -(a_n + [\lambda n + \mu] \times 2^n)$

则 $a_{n+1} = -a_n + (-3\lambda n - 3\mu - 2\lambda) \times 2^n$ 则 $\begin{cases} -3\lambda = 1 \\ -3\mu - 2\lambda = 0 \end{cases}$ 则 $\begin{cases} \lambda = -\frac{1}{3} \\ \mu = \frac{1}{9} \end{cases}$ 则 $a_{n+1} + (-\frac{1}{3}n + \frac{2}{9}) \times 2^{n+1} = (-1)^{n+1}[a_1 + (-\frac{1}{3} \times 1 + \frac{2}{9}) \times 2^1]$

则 $a_n = \frac{16}{9} \times (-1)^{n-1} + (\frac{1}{3}n - \frac{2}{9}) \times 2^n$

(14) 由 $a_{n+1} = \frac{a_n}{n+1}$, 则 $\frac{1}{a_{n+1}} = \frac{a_{n+1}}{a_n} = 1 + \frac{1}{a_n}$ 则 $\frac{1}{a_n} = \frac{1}{a_1} + (n-1) \times 1$ 则 $\frac{1}{a_n} = \frac{2n-1}{2}$ 则 $a_n = \frac{2}{2n-1}$

(15) 由 $a_{n+1} = \frac{a_n}{n+2}$, 则 $\frac{1}{a_{n+1}} = \frac{a_{n+1}}{a_n} = 1 + 2 \times \frac{1}{a_n}$ 设 $(\frac{1}{a_{n+1}} + \lambda) = 2(\frac{1}{a_n} + \lambda)$ 则 $\frac{1}{a_{n+1}} = \frac{2}{a_n} + \lambda$

则 $\lambda = 1$ 则 $\frac{1}{a_n} + 1 = 2^{n-1}(\frac{1}{a_1} + 1)$ 则 $\frac{1}{a_n} = 3 \times 2^{n-2} - 1$ 则 $a_n = \frac{1}{3 \times 2^{n-2} - 1}$

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(16) 由 $a_{n+1} = \frac{4a_n - 1}{2a_n + 1}$ 设 $a_{n+1} + \lambda = \frac{4a_n - 1}{2a_n + 1} + \lambda = \frac{(4+2\lambda)a_n + \lambda - 1}{2a_n + 1}$ 则 $\frac{1}{a_{n+1} + \lambda} = \frac{2a_n + 1}{(4+2\lambda)a_n + \lambda - 1} = \frac{\frac{2}{2\lambda+4}a_n + \frac{1}{2\lambda+4}}{a_n + \frac{\lambda-1}{2\lambda+4}}$

则 $\lambda = \frac{\lambda-1}{2\lambda+4}$ 则 $\lambda = -\frac{1}{2}$ 或 -1 当 $\lambda = -1$, 则 $\frac{1}{a_{n+1}-1} = \frac{a_n + \frac{1}{2}}{a_n - 1}$ ①; 当 $\lambda = -\frac{1}{2}$, 则 $\frac{1}{a_{n+1}-1} = \frac{2}{3} \times \frac{a_n + \frac{1}{2}}{a_n - \frac{1}{2}}$ ②

由 ① ÷ ② 则 $\frac{a_{n+1} - \frac{1}{2}}{a_{n+1} - 1} = \frac{3}{2} \times \frac{a_n - \frac{1}{2}}{a_n - 1}$ 则 $\frac{a_n - \frac{1}{2}}{a_n - 1} = (\frac{3}{2})^{n-1} (\frac{a_1 - \frac{1}{2}}{a_1 - 1}) = (\frac{3}{2})^n$ 则 $a_n - \frac{1}{2} = (\frac{3}{2})^n (a_1 - 1)$

则 $a_n = \frac{\frac{1}{2} - (\frac{3}{2})^n}{1 - (\frac{3}{2})^n} = \frac{3^n - 2^{n+1}}{3^n - 2^n}$

(17) 由 $a_{n+1} = \frac{a_n - 1}{a_n + 3}$, 设 $a_{n+1} + \lambda = \frac{a_n - 1}{a_n + 3} + \lambda = \frac{(\lambda+1)a_n + 3\lambda - 1}{a_n + 3}$ 则 $\frac{1}{a_{n+1} + \lambda} = \frac{a_n + 3}{(\lambda+1)a_n + 3\lambda - 1} = \frac{\frac{1}{\lambda+1}a_n + \frac{3}{\lambda+1}}{a_n + \frac{3\lambda-1}{\lambda+1}}$

则 $\lambda = \frac{3\lambda-1}{\lambda+1}$ 则 $\lambda = 1$ 则 $\frac{1}{a_{n+1}+1} = \frac{\frac{1}{2}a_n + \frac{3}{2}}{a_n + 1} = \frac{1}{a_n + 1} + \frac{1}{2}$ 则 $\frac{1}{a_{n+1}} = \frac{1}{a_1 + 1} + \frac{1}{2}(n-1) = \frac{3n-1}{6}$

则 $a_{n+1} = \frac{6}{3n-1}$ 则 $a_n = \frac{6}{3n-1} - 1 = \frac{7-3n}{3n-1}$

(18) 由 $a_{n+1} = a_n^2$, 则 $19a_{n+1} = 19a_n^2 = 219a_n$ 则 $19a_n = 2^{n-1} 19a_1 = (192) \times 2^{n-1}$ 则 $a_n = 10^{(192) \times 2^{n-1}}$
 $= (10^{192})^{2^{n-1}} = 2^{2^{n-1}}$

(19) 由 $a_{n+2} = 5a_{n+1} - 6a_n$, 设 $a_{n+2} + \lambda a_{n+1} = (5+\lambda)(a_{n+1} + \lambda a_n)$ 则 $a_{n+2} = 5a_{n+1} + \lambda(5+\lambda)a_n$

则 $(5+\lambda)\lambda = -6$ 则 $\lambda = -3$ 或 -2 不妨取 $\lambda = -2$ 则 $a_{n+2} - 2a_{n+1} = 3(a_{n+1} - 2a_n)$

则 $a_{n+1} - 2a_n = 3^{n-1}(a_2 - 2a_1)$ 则 $a_{n+1} - 2a_n = -3^{n-1}$ 则 $a_{n+1} = 2a_n - 3^{n-1}$ 设 $a_{n+1} + \lambda \cdot 3^{n-1} = 2(a_n + \lambda \cdot 3^{n-1})$

则 $a_{n+1} = 2a_n - 3\lambda \times 3^{n-1}$ 则 $-3\lambda = -1$ 则 $\lambda = \frac{1}{3}$ 则 $a_n + 3^{n-1} = 2^{n-1}(a_1 + 3^{1-1}) = 3 \times 2^{n-1}$

则 $a_n = 3 \times 2^{n-1} - 3^{n-1}$

(20) 由 $a_{n+2} = 4a_{n+1} - 4a_n$, 设 $a_{n+2} + \lambda a_{n+1} = (4+\lambda)(a_{n+1} + \lambda a_n)$ 则 $a_{n+2} = 4a_{n+1} + \lambda(\lambda+4)a_n$

则 $(\lambda+4)\lambda = -4$ 则 $\lambda = -2$ 则 $a_{n+2} - 2a_{n+1} = 2(a_{n+1} - 2a_n)$ 则 $a_{n+1} - 2a_n = 2^{n-1}(a_2 - 2a_1)$

则 $a_{n+1} - 2a_n = -2^{n-1}$ 则 $a_{n+1} = 2a_n - 2^{n-1}$ 同除 2^{n+1} 则 $\frac{a_{n+1}}{2^{n+1}} = \frac{a_n}{2^n} - \frac{1}{4}$ 则 $\frac{a_n}{2^n} = \frac{a_1}{2^1} - \frac{1}{4}(n-1)$

则 $\frac{a_n}{2^n} = \frac{5-n}{4}$ 则 $a_n = (5-n) \times 2^{n-2}$

(21) 由 $a_{n+1} = \begin{cases} 2a_n, & n \text{ 为奇数} \\ a_{n+1}, & n \text{ 为偶数} \end{cases}$ 当 n 为偶数时, 则 $n+1$ 为奇数, 则 $a_{n+1} = a_n + 1, a_{n+2} = 2a_{n+1}$

则 $a_{n+2} = 2a_{n+1} = 2(a_n + 1) = 2a_n + 2$ 则 $a_{n+2} + 2 = 2(a_n + 2)$ 则 $a_n + 2 = 2^{\frac{n+2}{2}-1}(a_2 + 2) = 2^{\frac{n+2}{2}}$

则 $a_n = 2^{\frac{n+2}{2}} - 2$ (n 为偶数) 当 n 为奇数时, 则 $n+1$ 为偶数, 则 $a_{n+1} = 2a_n$

则 $a_n = \frac{1}{2} \cdot a_{n+1} = \frac{1}{2}(2^{\frac{n+3}{2}} - 2) = 2^{\frac{n+1}{2}} - 1$ 综上 $a_n = \begin{cases} 2^{\frac{n+2}{2}} - 1, & n \text{ 为奇数} \\ 2^{\frac{n+2}{2}} - 2, & n \text{ 为偶数} \end{cases}$

(22) 当 n 为偶数时, 则 $n+1$ 为奇数, 则 $a_{n+1} + a_n = 2n+1, a_{n+2} - a_{n+1} = 2(n+1)$,

则 $a_{n+2} = a_{n+1} + 2(n+1) = (2n - a_n) + 2(n+1) = -a_n + 4n + 2$ 设 $a_{n+2} + \lambda(n+2) + \mu = -(a_n + \lambda n + \mu)$

则 $a_{n+2} = -a_n - 2\lambda n - 2\mu - 2\lambda$ 则 $\begin{cases} -2\lambda = 4 \\ -2\mu - 2\lambda = 2 \end{cases}$ 则 $\begin{cases} \lambda = -2 \\ \mu = 1 \end{cases}$ 则 $a_n - 2n + 1 = (-1)^{\frac{n}{2}-1}(a_2 - 2 \times 2 + 1)$

由 $a_2 - a_1 = 2 \times 1$, 则 $a_2 = 4$ 则 $a_n = (-1)^{\frac{n}{2}-1} + 2n - 1$ (n 为偶数)

当 n 为奇数时, 则 $n+1$ 为偶数, 则 $a_{n+1} - a_n = 2n$ 则 $a_n = a_{n+1} - 2n = [(-1)^{\frac{n+1}{2}-1} + 2(n+1) - 1] - 2n = (-1)^{\frac{n}{2}} + 1$ (不妨知)

综上 $a_n = \begin{cases} (-1)^{\frac{n}{2}-1} + 2n - 1, & n \text{ 为偶数} \\ (-1)^{\frac{n}{2}} + 1, & n \text{ 为奇数} \end{cases}$

(23) 当 n 为偶数时, 则 $n+1$ 为奇数, 则 $a_{n+1}a_n = 4n^2 - 1, a_{n+2}a_{n+1} = 4(n+1)^2 - 1$, 则 $\frac{a_{n+2}}{a_n} = \frac{2n+3}{2n-1} = \frac{2(n+2)-1}{2n-1}$

则 $\frac{a_n}{a_2} = \frac{2n-1}{2 \times 2 - 1} = \frac{2n-1}{3}$ 又由 $a_1a_2 = 4 \times 1^2 - 1 = 3$, 则 $a_2 = \frac{3}{2}$ 则 $a_n = \frac{2n-1}{2} = n - \frac{1}{2}$ (n 为偶)

当 n 为奇数时, 则 $n+1$ 为偶数, 则 $a_n a_{n+1} = 4n^2 - 1$, 则 $a_n = \frac{4n^2 - 1}{a_{n+1}} = (4n^2 - 1) \cdot \frac{1}{2(n+1)-1} = 2(2n-1) = 4n-2$ (n 为奇)

则 $a_n = \begin{cases} n - \frac{1}{2}, & n \text{ 为偶数} \\ 4n - 2, & n \text{ 为奇数} \end{cases}$

例2: (1) 由 $a_n = \frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$, 则 $S_n = a_1 + a_2 + \dots + a_n = \frac{1}{1} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \dots - \frac{1}{n} + \frac{1}{n+1} = 1 - \frac{1}{n+1} = \frac{n}{n+1}$

(2) 由 $a_n = \frac{1}{n(n+2)} = \frac{1}{2}(\frac{1}{n} - \frac{1}{n+2})$, 则 $S_n = \frac{1}{2}(\frac{1}{1} - \frac{1}{3} + \frac{1}{2} - \frac{1}{4} + \dots - \frac{1}{n-1} + \frac{1}{n+1} - \frac{1}{n+2}) = \frac{1}{2}(1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2}) = \frac{3}{4} - \frac{1}{2}(\frac{1}{n+1} + \frac{1}{n+2})$

(3) 由 $a_n = \frac{1}{4n^2-1} = \frac{1}{(2n-1)(2n+1)} = \frac{1}{2}(\frac{1}{2n-1} - \frac{1}{2n+1})$, 则 $S_n = \frac{1}{2}(\frac{1}{1} - \frac{1}{3} + \dots - \frac{1}{2n-1} + \frac{1}{2n+1}) = \frac{1}{2}(1 - \frac{1}{2n+1}) = \frac{n}{2n+1}$

(4) 由 $a_n = \frac{n^2}{4n^2-1} = \frac{\frac{1}{4}(4n^2-1) + \frac{3}{4}}{4n^2-1} = \frac{1}{4} + \frac{3}{4} \cdot \frac{1}{4n^2-1} = \frac{1}{4} + \frac{3}{8}(\frac{1}{2n-1} - \frac{1}{2n+1})$, 则 $S_n = \frac{n}{4} + \frac{3}{8}(\frac{1}{1} - \frac{1}{3} + \dots - \frac{1}{2n-1} + \frac{1}{2n+1}) = \frac{n}{4} + \frac{1}{8n+4}$

(5) 由 $a_n = \frac{1}{\sqrt{n+1} + \sqrt{n}} = \sqrt{n+1} - \sqrt{n}$, 则 $S_n = (\sqrt{2} - \sqrt{1} + \dots - \sqrt{n+1} + \sqrt{n}) = \sqrt{n+1} - 1$

(6) 由 $a_n = \frac{1}{(n+1)\sqrt{n} + n\sqrt{n+1}} = \frac{1}{\sqrt{n(n+1)}(\sqrt{n+1} + \sqrt{n})} = \frac{\sqrt{n+1} - \sqrt{n}}{\sqrt{n(n+1)}} = \frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}}$, 则 $S_n = (\frac{1}{\sqrt{1}} - \frac{1}{\sqrt{2}} + \dots - \frac{1}{\sqrt{n}} + \frac{1}{\sqrt{n+1}}) = 1 - \frac{1}{\sqrt{n+1}}$

(7) 由 $a_n = \sqrt{1 + \frac{1}{n^2} + \frac{1}{(n+1)^2}} = \sqrt{\frac{1}{n^2} + \frac{1}{(n+1)^2} - \frac{2}{n(n+1)} + \frac{2}{n(n+1)} + 1} = \sqrt{(\frac{1}{n} - \frac{1}{n+1})^2 + \frac{2}{n(n+1)} + 1} = \sqrt{(\frac{1}{n(n+1)})^2 + \frac{2}{n(n+1)} + 1} = \sqrt{(\frac{1}{n(n+1)} + 1)^2} = \frac{1}{n(n+1)} + 1 = 1 + \frac{1}{n} - \frac{1}{n+1}$, 则 $S_n = n + (\frac{1}{1} - \frac{1}{2} + \dots - \frac{1}{n} + \frac{1}{n+1}) = n + 1 - \frac{1}{n+1} = \frac{n(n+2)}{n+1}$

(8) 由 $a_n = \frac{1}{n(n+1)(n+2)} = \frac{n+1}{n(n+1) \cdot (n+1)(n+2)} = \frac{1}{2}[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}]$, 则 $S_n = \frac{1}{2}(\frac{1}{1 \cdot 2} - \frac{1}{2 \cdot 3} + \dots - \frac{1}{n(n+1)} + \frac{1}{(n+1)(n+2)}) = \frac{1}{2}(\frac{1}{2} - \frac{1}{(n+1)(n+2)}) = \frac{1}{4} - \frac{1}{2(n+1)(n+2)}$ (注: 设 $\frac{A}{n(n+1)} - \frac{A}{(n+1)(n+2)} = \frac{2A(n+1)}{n(n+1)^2(n+2)} = \frac{2A}{n(n+1)(n+2)}$, 则 $A = \frac{1}{2}$)

(9) 由 $a_n = \frac{2 \times 3^n}{(3^{n+1} + 1)(3^{n+1} - 1)} = \frac{3^{n+1} - (3^n + 1)}{(3^{n+1} + 1)(3^{n+1} - 1)} = \frac{1}{3^{n+1} + 1} - \frac{1}{3^{n+1} - 1}$, 则 $S_n = (\frac{1}{3+1} - \frac{1}{3^2+1} + \dots - \frac{1}{3^{n+1}+1} + \frac{1}{3^{n+1}-1}) = \frac{1}{4} - \frac{1}{3^{n+1}-1}$ (注: $\frac{A}{3^{n+1}-1} - \frac{A}{3^{n+1}+1} = \frac{2A \times 3^n}{(3^{n+1}+1)(3^{n+1}-1)}$, 则 $A = 1$)

(10) 由 $a_n = \frac{n+2}{n(n+1) \cdot 2^n} = \frac{(n+2)2^{n+1}}{n \cdot 2^n \cdot (n+1) \cdot 2^{n+1}} = 2 \times [\frac{(n+1)2^{n+1} - n2^n}{n \cdot 2^n \cdot (n+1) \cdot 2^{n+1}}] = 2(\frac{1}{n \cdot 2^n} - \frac{1}{(n+1)2^{n+1}}) = \frac{1}{n \cdot 2^{n-1}} - \frac{1}{(n+1)2^n}$

则 $S_n = (\frac{1}{1 \cdot 2^{0-1}} - \frac{1}{2 \cdot 2^1} + \dots - \frac{1}{n \cdot 2^{n-1}} + \frac{1}{(n+1)2^n}) = 1 - \frac{1}{(n+1)2^n}$ (注: $\frac{A}{n \cdot 2^n} - \frac{A}{(n+1)2^{n+1}} = \frac{A(n+2)2^n}{n(n+1)2^n \cdot 2^{n+1}} = \frac{\frac{1}{2}A(n+2)}{n(n+1)2^n}$, 则 $\frac{1}{2}A = 1$, 则 $A = 2$)

(11) 由 $a_n = \frac{n \cdot 2^n}{(n+1)(n+2)} = \frac{(n+1)2^{n+1} - (n+2)2^n}{(n+1)(n+2)} = \frac{2^{n+1}}{n+2} - \frac{2^n}{n+1}$, 则 $S_n = (\frac{2^{n+1}}{1+2} - \frac{2^1}{1+1} + \dots - \frac{2^{n+1}}{n+2} + \frac{2^n}{n+1}) = \frac{2^{n+1}}{n+2} - 1$ (注: $\frac{A \cdot 2^n}{n+1} - \frac{A \cdot 2^{n+1}}{n+2} = \frac{-A \cdot 2^n}{(n+1)(n+2)}$, 则 $-A = 1$, 则 $A = -1$)

(12) 由 $a_n = \frac{2n+1}{n^2(n+1)^2} = \frac{(n+1)^2 - n^2}{n^2(n+1)^2} = \frac{1}{n^2} - \frac{1}{(n+1)^2}$, 则 $S_n = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{2^2} - \frac{1}{3^2} + \dots - \frac{1}{n^2} + \frac{1}{(n+1)^2} = 1 - \frac{1}{(n+1)^2}$

(注: $\frac{A}{n^2} - \frac{A}{(n+1)^2} = \frac{A(2n+1)}{n^2(n+1)^2}$, 则 $A = 1$)

(13) 由 $a_n = (-1)^n(4n-3)$, 设 $b_n = 4n-3$, 当 n 为偶数, 则 $S_n = -b_1 + b_2 - b_3 + b_4 + \dots + (-1)b_{n-1} + b_n = \frac{n}{2}(b_n - b_{n-1}) = \frac{n}{2} \times 4 = 2n$ 当 n 为奇数, 则 $n+1$ 为偶数, 则 $S_n = S_{n+1} - a_{n+1} = 2(n+1) - [4(n+1)-3] = 1-2n$ 综上 $S_n = \begin{cases} 2n, & n \text{ 为偶数} \\ 1-2n, & n \text{ 为奇数} \end{cases}$

(14) 由 $a_n = (-1)^n(4n-3)$, 设 $b_n = 4n-3$, 当 n 为偶数, 则 $S_n = -b_1^2 + b_2^2 - b_3^2 + b_4^2 + \dots + (-1)b_{n-1}^2 + b_n^2 = (b_2 - b_1)(b_2 + b_1) + (b_4 - b_3)(b_4 + b_3) + \dots + (b_n - b_{n-1})(b_n + b_{n-1}) = 4 \times (b_1 + b_2 + \dots + b_n) = 4 \times \frac{(1+4n-3)n}{2} = 4n(2n-1)$ 当 n 为奇数, 则 $S_n = S_{n+1} - a_{n+1} = 4(n+1)[2(n+1)-1] - [4(n+1)-3] = 8(n+1)^2 - 8(n+1) + 3$

(15) 由 $a_n = n \cdot 2^n$ 设 $n \cdot 2^n = [A(n+1) + B] \times 2^{n+1} - [A(n+1) + B] \times 2^n = (A(n+1) + 2A + B) \times 2^n$ 则 $\begin{cases} A=1 \\ 2A+B=0 \end{cases}$ 则 $\begin{cases} A=1 \\ B=-2 \end{cases}$ 则 $S_n = 1 \times 2^1 + 2 \times 2^2 + \dots + n \times 2^n = (1-1) \times 2^{1+1} - (1-2) \times 2^1 + \dots + (n-1) \times 2^{n+1} - (n-2) \times 2^n = (n-1) \times 2^{n+1} + 2$

(16) 由 $a_n = \frac{n}{2^n}$ 设 $\frac{n}{2^n} = \frac{A(n+1)}{2^n} - \frac{A(n+1)+B}{2^{n+1}} = \frac{\frac{1}{2}A(n+1)-\frac{1}{2}A}{2^n}$ 则 $\begin{cases} \frac{1}{2}A=1 \\ \frac{1}{2}B-\frac{1}{2}A=0 \end{cases}$ 则 $\begin{cases} A=2 \\ B=2 \end{cases}$ 则 $S_n = \frac{2 \times 1 + 2}{2^1} - \frac{2 \times (1+1) + 2}{2^2} + \dots - \frac{2n+2}{2^n} + \frac{2(n+1)+2}{2^{n+1}} = 2 - \frac{n+2}{2^n}$

(17) 由 $\cos(\frac{n\pi}{2})$ 周期 $T = \frac{2\pi}{\frac{\pi}{2}} = 4$, 则把每四项看作一个整体 由 $a_n = n\cos(\frac{n\pi}{2}) + 1$

设 $n = 4k, n = 4k-1, n = 4k-2, n = 4k-3$ 其中 $k \in \mathbb{N}^*$

$$\text{则 } a_{4k} = 4k\cos(\frac{4k\pi}{2}) + 1 = 4k + 1; a_{4k-1} = (4k-1)\cos(\frac{4k-1}{2}\pi) + 1 = 4k\pi\cos(2k\pi - \frac{\pi}{2}) + 1 = 1;$$

$$a_{4k-2} = (4k-2)\cos(\frac{4k-2}{2}\pi) + 1 = (4k-2)\cos(2k\pi - \pi) + 1 = -4k+3; a_{4k-3} = (4k-3)\cos(\frac{4k-3}{2}\pi) + 1 = (4k-3)\cos(2k\pi - \frac{\pi}{2}) + 1 = 1$$

$$\text{当 } n = 4k, \text{ 由 } a_{4k-3} + a_{4k-2} + a_{4k-1} + a_{4k} = 1 + (-4k+3) + 1 + 4k + 1 = 6$$

$$\text{则 } S_n = S_{4k} = (a_1 + a_2 + a_3 + a_4) + \dots + (a_{4k-3} + a_{4k-2} + a_{4k-1} + a_{4k}) = 6k$$

$$\text{当 } n = 4k-1, \text{ 由 } S_n = S_{4k} - a_{4k} = 6k - (4k+1) = 2k-1$$

$$\text{当 } n = 4k-2, \text{ 由 } S_n = S_{4k-1} - a_{4k-1} = 2k-1 - 1 = 2k-2$$

$$\text{当 } n = 4k-3, \text{ 由 } S_n = S_{4k-2} - a_{4k-2} = 2k-2 - (-4k+3) = 6k-5$$

$$\text{综上 } S_n = \begin{cases} 6k-5, & n=4k-3 \\ 2k-2, & n=4k-2 \\ 2k-1, & n=4k-1 \\ 6k, & n=4k \end{cases} \text{ 其中 } k \in \mathbb{N}^*$$

(18) 由 $a_n = |23-7n|$ 设 $b_n = 23-7n$, 令 $b_n \geq 0$, 则 $n \leq \frac{23}{7}$ 又由 $n \in \mathbb{N}^*$, 则 $n \leq 3$

当 $n \leq 3$ 且 $n \in \mathbb{N}^*$ 时, $b_n \geq 0$; 当 $n > 3$ 且 $n \in \mathbb{N}^*$ 时, $b_n < 0$

$$\text{当 } n \leq 3 \text{ 且 } n \in \mathbb{N}^* \text{ 时, } S_n = b_1 + \dots + b_n = \frac{(16+23-7n) \times n}{2} = \frac{(39-7n)n}{2}; \text{ 当 } n > 3 \text{ 时, } S_n = b_1 + b_2 + b_3 -$$

$$-(b_4 + b_5 + \dots + b_n) = -(b_1 + b_2 + \dots + b_n) + 2(b_1 + b_2 + b_3) = -\frac{(39-7n)n}{2} + 2 \times \frac{(39-7 \times 3) \times 3}{2} = 54 + \frac{(7n-39)n}{2}$$

$$(19) \text{ 由 } a_n = \begin{cases} 2^n, & n \text{ 为奇数} \\ 2^n, & n \text{ 为偶数} \end{cases} \text{ 当 } n \text{ 为偶数时, } S_n = (a_1 + a_3 + \dots + a_{n-1}) + (a_2 + a_4 + \dots + a_n) = \frac{[2 \times (1+2^{n-1})] \times \frac{n}{2}}{2} + \frac{2^2[(2^{\frac{n}{2}}-1)]}{2^2-1} = \frac{1}{2}n^2 + \frac{4}{3}(2^n-1)$$

$$\text{当 } n \text{ 为奇数时, } S_n = S_{n+1} - a_{n+1} = \frac{1}{2}(n+1)^2 + \frac{4}{3}(2^{n+1}-1) - 2^{n+1} = \frac{1}{2}(n+1)^2 + \frac{4}{3}(2^{n+1}-4)$$

$$\text{综上 } S_n = \begin{cases} \frac{1}{2}n^2 + \frac{4}{3}(2^n-1), & n \text{ 为偶数} \\ \frac{1}{2}(n+1)^2 + \frac{4}{3}(2^{n+1}-4), & n \text{ 为奇数} \end{cases}$$

例3: (1) 由 $4S_n = (a_n+1)^2$ ① 则 $4S_{n+1} = (a_{n+1}+1)^2$ ② 由 ②-① 则 $4a_{n+1} = (a_{n+1}+1)^2 - (a_n+1)^2$

$$\text{则 } (a_n+1)^2 = (a_{n+1}+1)^2 - 4a_{n+1} = (a_{n+1}-1)^2 \text{ 则 } a_n+1 = a_{n+1}-1 \text{ 或 } a_n+1 = 1-a_{n+1}$$

若 $a_n+1 = 1-a_{n+1}$ 则 $a_{n+1} = -a_n$ 与 $\{a_n\}$ 为正项数列不符, 舍.

若 $a_n+1 = a_{n+1}-1$ 则 $a_{n+1} - a_n = 2$ 又由 $4S_1 = (a_1+1)^2$ 则 $(a_1-1)^2 = 0$ 则 $a_1 = 1$.

$$\text{则 } a_n = 1 + 2(n-1) = 2n-1$$

(2) 由 $4S_n = (a_{n+1}-1)^2$ ①, 则 $4S_{n+1} = (a_{n+2}-1)^2$ ② 由 ②-① 得, $4a_{n+1} = (a_{n+2}-1)^2 - (a_{n+1}-1)^2$

$$\text{则 } (a_{n+2}-1)^2 = (a_{n+1}-1)^2 + 4a_{n+1} = (a_{n+1}+1)^2 \text{ 则 } a_{n+2}-1 = a_{n+1}+1 \text{ 或 } a_{n+2}-1 = -1-a_{n+1}$$

若 $a_{n+2}-1 = -1-a_{n+1}$ 则 $a_{n+2} = -a_{n+1}$ 这与 $\{a_n\}$ 为正项数列不符, 舍

若 $a_{n+2}-1 = a_{n+1}+1$ 则 $a_{n+2} - a_{n+1} = 2$ 当 $n=1$ 时, $4S_1 = (a_2-1)^2$ 则 $a_2 = 3$. 则 $a_2 - a_1 = 3-1=2$

$$\text{则 } a_{n+1} - a_n = 2 \text{ 又由 } a_1 = 1 \text{ 则 } a_n = 1 + 2(n-1) = 2n-1$$

(3) 由 $4S_2 = (a_1+3)^2$ 又由 $S_2 = a_1 + a_2, a_1 = 1$ 则 $4(a_1+a_2) = (a_1+3)^2$ 则 $4a_2 = (a_1+3)^2 - 4a_1$

$$\text{则 } a_2 = 3 \text{ 同理 } 4S_3 = (a_2+3)^2, \text{ 则 } 4a_3 = (a_2+3)^2 - 4(a_1+a_2) \text{ 则 } a_3 = 5$$

猜想 $a_n = 2n-1$ 为通项公式. 下证: $a_n = 2n-1$ 为所求.

当 $n=1$ 时, $a_1=1$ 符合上式. 假设 $n=k (k \in \mathbb{N}^*)$, $a_k = 2k-1$ 成立. 则 $S_k = \frac{(1+2k-1)k}{2} = k^2$

$$\text{由 } 4S_{k+1} = (a_{k+1}+3)^2 \quad \text{则 } 4a_{k+1} = (a_k+3)^2 - 4S_k = (2k-1+3)^2 - 4k^2 = 8k+4$$

则 $a_{k+1} = 2k+1 = 2(k+1)-1$ 则当 $n=k$ 时, $n=k+1$ 也成立.

综上所述, $a_n = 2n-1$ 为所求.

(4) 法一: 当 $n=1$ 时, $S_1 = \frac{a_1}{2} + \frac{1}{a_1} - 1$, 则 $a_1 = \sqrt{3} - 1$ 当 $n \geq 2$ 时, $S_n = \frac{S_n - S_{n-1}}{2} + \frac{1}{S_n - S_{n-1}} - 1$

$$\text{则 } 2S_n(S_n - S_{n-1}) = (S_n - S_{n-1})^2 + 2 - 2(S_n - S_{n-1}) \quad \text{则 } S_n^2 + 2S_n + 1 = S_{n-1}^2 + 2S_{n-1} + 1 + 2$$

$$\text{则 } (S_n + 1)^2 - (S_{n-1} + 1)^2 = 2 \quad \text{则 } (S_n + 1)^2 = (S_1 + 1)^2 + 2(n-1) = 2n+1 \quad \text{则 } S_n = \sqrt{2n+1} - 1$$

当 $n \geq 2$ 时, $a_n = S_n - S_{n-1} = \sqrt{2n+1} - \sqrt{2n-1}$ 当 $n=1$ 时, $a_1 = \sqrt{3} - 1$ 符合左式. 综上 $a_n = \sqrt{2n+1} - \sqrt{2n-1}$

法二: 当 $n=1$ 时, $S_1 = \frac{a_1}{2} + \frac{1}{a_1} - 1$, 则 $a_1 = \sqrt{3} - 1$, 当 $n=2$ 时, $S_2 = \frac{a_2}{2} + \frac{1}{a_2} - 1$ 则 $a_2 = \sqrt{5} - \sqrt{3}$

当 $n=3$ 时, $S_3 = \frac{a_3}{2} + \frac{1}{a_3} - 1$, 则 $a_3 = \sqrt{7} - \sqrt{5}$ 猜想 $a_n = \sqrt{2n+1} - \sqrt{2n-1}$

假设 $n=k (k \in \mathbb{N}^*)$ 时成立, $a_k = \sqrt{2k+1} - \sqrt{2k-1}$, 则 $S_k = \sqrt{2k+1} - 1$ 由 $S_{k+1} = \frac{a_{k+1}}{2} + \frac{1}{a_{k+1}} - 1$

$$\text{则 } S_k + a_{k+1} = \frac{a_{k+1}}{2} + \frac{1}{a_{k+1}} - 1 \quad \text{则 } a_{k+1}^2 + 2\sqrt{2k+1}a_{k+1} - 2 = 0 \quad \text{则 } a_{k+1} = \frac{-2\sqrt{2k+1} \pm \sqrt{4 \times (2k+1) - 4 \times (-2)}}{2}$$

$$= \frac{-2\sqrt{2k+1} \pm 2\sqrt{2k+3}}{2} \quad \text{由 } a_n > 0 \quad \text{则 } a_{k+1} = \sqrt{2k+3} - \sqrt{2k+1} = \sqrt{2(k+1)+1} - \sqrt{2(k+1)-1}$$

则 $n=k$ 成立时, $n=k+1$ 也成立. 所以 $a_n = \sqrt{2n+1} - \sqrt{2n-1}$ 为所求.

(5) 法一: 由 $2S_n + 1 = 2a_n^2 + a_n$ ① 则 $2S_{n+1} + 1 = 2a_{n+1}^2 + a_{n+1}$ ② 由 ② - ① 则 $2a_{n+1} = 2a_n^2 - 2a_n + a_{n+1} - a_n$

$$\text{则 } a_{n+1} + a_n = 2(a_n^2 - a_n) \quad \text{则 } a_{n+1} - a_n = \frac{1}{2} \quad \text{又由 } 2S_1 + 1 = 2a_1^2 + a_1 \quad \text{则 } a_1 = 1 \text{ 或 } -\frac{1}{2} \text{ (舍)}.$$

$$\text{则 } a_n = 1 + \frac{1}{2}(n-1) = \frac{n+1}{2}$$

法二: 由 $2S_1 + 1 = 2a_1^2 + a_1$, 则 $a_1 = 1$; 又由 $2S_2 + 1 = 2a_2^2 + a_2$ 则 $a_2 = \frac{3}{2}$ 又由 $2S_3 + 1 = 2a_3^2 + a_3$

则 $a_3 = 2$ 猜想 $a_n = \frac{n+1}{2}$ 为所求. 假设 $n=k (k \in \mathbb{N}^*)$ 时成立, 则 $a_k = \frac{k+1}{2}$ 则 $S_k = \frac{(1+\frac{k+1}{2})k}{2}$

$$= \frac{k(k+3)}{4} \quad \text{由 } 2S_{k+1} + 1 = 2a_{k+1}^2 + a_{k+1} \quad \text{则 } 2S_k + 2a_{k+1} + 1 = 2a_{k+1}^2 + a_{k+1}$$

$$\text{则 } 2a_{k+1}^2 - a_{k+1} - 2S_k - 1 = 0 \quad \text{则 } 2a_{k+1}^2 - a_{k+1} - \frac{1}{2}(k+1)(k+2) = 0 \quad \text{则 } a_{k+1} = \frac{1 \pm \sqrt{1 - 4 \times (-\frac{1}{2})(k+1)(k+2)}}{2 \times 2}$$

$$= \frac{1 \pm \sqrt{(2k+3)^2}}{4} = \frac{1 \pm (2k+3)}{4} \quad \text{由 } a_n > 0 \quad \text{则 } a_{k+1} = \frac{1+2k+3}{4} = \frac{2k+4}{4} = \frac{k+2}{2} = \frac{(k+1)+1}{2}$$

则 $n=k$ 成立时, $n=k+1$ 也成立. 所以 $a_n = \frac{n+1}{2}$ 为所求.

(6) 由 $\frac{S_{n+1}}{n+1} - \frac{S_n}{n} = 1$, 则 $\frac{S_n}{n} = \frac{S_1}{1} + 1 \times (n-1) = n$ 则 $S_n = n^2$ 当 $n \geq 2$ 时, $a_n = S_n - S_{n-1} = n^2 - (n-1)^2 = 2n-1$

当 $n=1$ 时, $a_1=1$ 符合上式. 则 $a_n = 2n-1$ 为所求.

(7) 由 $a_n = \frac{S_n}{n} + n-1$, 当 $n \geq 2$ 时, $S_n - S_{n-1} = \frac{S_n}{n} + n-1$ 则 $S_n(1 - \frac{1}{n}) = S_{n-1} + n-1$ 则 $S_n \frac{n-1}{n} = S_{n-1} + n-1$

则 $\frac{S_n}{n} = \frac{S_{n-1}}{n-1} + 1$ 则 $\frac{S_n}{n} = \frac{S_1}{1} + 1 \times (n-1) = n$ 则 $S_n = n^2$ 当 $n \geq 2$ 时, $a_n = S_n - S_{n-1} = n^2 - (n-1)^2 = 2n-1$

当 $n=1$ 时, $a_1=1$ 也符合左式. 则 $a_n = 2n-1$ 为所求.

(8) 由 $\sqrt{S_{n+1}} - \sqrt{S_n} = 1$ 则 $\sqrt{S_n} = \sqrt{a_1} + (n-1) \times 1 = n$ 则 $S_n = n^2$ 当 $n \geq 2$ 时, $a_n = S_n - S_{n-1} = n^2 - (n-1)^2 = 2n-1$

当 $n=1$ 时, $a_1=1$ 也符合左式. 则 $a_n = 2n-1$ 为所求.

(9) 由 $a_{n+1}(S_{n+1}+S_n)=1$ 则 $(S_{n+1}-S_n)(S_{n+1}+S_n)=1$ 则 $S_{n+1}^2-S_n^2=1$ 则 $S_n^2=S_1^2+(n-1)\times 1=n$

则 $S_n=\sqrt{n}$ 当 $n\geq 2$ 时, $a_n=S_n-S_{n-1}=\sqrt{n}-\sqrt{n-1}$ 当 $n=1$ 时, $a_1=1$ 符合左式 则 $a_n=\sqrt{n}-\sqrt{n-1}$ 为所求

(10) 法一: 由 $\frac{S_n}{a_{n+1}}-\frac{S_n}{a_n}=-\frac{1}{2}$ 则 $\frac{S_{n+1}-a_{n+1}}{a_{n+1}}-\frac{S_n}{a_n}=-\frac{1}{2}$ 则 $\frac{S_{n+1}}{a_{n+1}}-\frac{S_n}{a_n}=\frac{1}{2}$ 则 $\frac{S_n}{a_n}=\frac{S_1}{a_1}+\frac{1}{2}(n-1)=\frac{n+1}{2}$

则 $S_n=\frac{n+1}{2}a_n$ ① 则 $S_{n+1}=\frac{n+2}{2}a_{n+1}$ ② 由 ②-①, 则 $a_{n+1}=\frac{n+2}{2}a_{n+1}-\frac{n+1}{2}a_n$ 则 $a_{n+1}\frac{n}{2}=\frac{n+1}{2}a_n$

则 $\frac{a_{n+1}}{a_n}=\frac{n+1}{n}$ 则 $\frac{a_n}{a_1}=\frac{n}{1}$ 则 $a_n=n$

法二: 当 $n=1$ 时, $\frac{S_1}{a_2}-\frac{S_1}{a_1}=-\frac{1}{2}$ 则 $a_2=2$ 当 $n=2$ 时, $\frac{S_2}{a_3}-\frac{S_2}{a_2}=-\frac{1}{2}$ 则 $a_3=3$

猜想 $a_n=n$ 为所求 假设 $n=k(k\in\mathbb{N}^*)$ 时成立, 则 $a_k=k$, 则 $S_k=\frac{k(k+1)}{2}$

由 $\frac{S_k}{a_{k+1}}-\frac{S_k}{a_k}=-\frac{1}{2}$ 则 $\frac{1}{a_{k+1}}=\frac{1}{a_k}-\frac{1}{2S_k}=\frac{1}{k}-\frac{1}{k(k+1)}=\frac{1}{k}-\left(\frac{1}{k}-\frac{1}{k+1}\right)=\frac{1}{k+1}$ 则 $a_{k+1}=k+1$

当 $n=k$ 成立时, $n=k+1$ 也成立. 综上, $a_n=n$ 为所求.

(11) 法一: 由 $S_{n+1}+S_n=(n+1)^2$, 则 $S_{n+1}=-S_n+(n+1)^2=-S_n+n^2+2n+1$ 设 $S_{n+1}=A(n+1)^2+B(n+1)+C$
 $=-(S_n+An^2+Bn+C)$ 则 $S_{n+1}=-S_n-2An^2+(-2A-2B)n-2C-A-B$ 则 $\begin{cases} -2A=1 \\ -2A-2B=2 \\ -2C-A-B=1 \end{cases}$ 则 $\begin{cases} A=-\frac{1}{2} \\ B=-\frac{1}{2} \\ C=0 \end{cases}$

则 $S_n-\frac{1}{2}n^2-\frac{1}{2}n=(-1)^{n-1}(S_1-\frac{1}{2}\times 1^2-\frac{1}{2}\times 1)$ 则 $S_n=\frac{n(n+1)}{2}$ 当 $n\geq 2$ 时, $a_n=S_n-S_{n-1}=\frac{n(n+1)}{2}-\frac{n(n-1)}{2}$

$=n$ 当 $n=1$ 时, $a_1=1$ 符合左式 则 $a_n=n$ 为所求

法二: 当 $n=1$ 时, $S_2+S_1=(1+1)^2$, 则 $a_2=2$; 当 $n=2$ 时, $S_3+S_2=(2+1)^2$, 则 $a_3=3$

猜想 $a_n=n$ 为所求, 假设 $n=k(k\in\mathbb{N}^*)$ 时成立, 则 $a_k=k$, $S_k=\frac{k(k+1)}{2}$

由 $S_{k+1}=-S_k+(k+1)^2$ 则 $a_{k+1}+S_k=-S_k+(k+1)^2$ 则 $a_{k+1}=-2S_k+(k+1)^2=-k(k+1)+k+1^2=k+1$ 则当 $n=k$ 成立时, $n=k+1$ 也成立. 所以 $a_n=n$ 为所求

(12) 法一: 由 $a_1^3+\dots+a_n^3=S_n^2$ ① 则 $a_1^3+\dots+a_{n+1}^3=S_{n+1}^2$ ② 由 ②-① 则 $a_{n+1}^3=S_{n+1}^2-S_n^2$

则 $(S_{n+1}-S_n)^3=S_{n+1}^2-S_n^2$ 则 $(S_{n+1}-S_n)^2=S_{n+1}+S_n$ 则 $a_{n+1}^2=S_{n+1}+S_n$ ① 则

$a_{n+2}^2=S_{n+2}+S_{n+1}$ ② 由 ②-① 则 $a_{n+2}^2-a_{n+1}^2=S_{n+2}-S_{n+1}+S_{n+1}-S_n=a_{n+2}+a_{n+1}$

则 $a_{n+2}-a_{n+1}=1$ 又由 $n=1$ 时, $a_1^3=S_1^2$ 则 $a_1=1$ 或 0 (舍), $n=2$ 时, $a_1^3+a_2^3=S_2^2$

则 $a_2=2$ 则 $a_2-a_1=1$ 则 $a_{n+1}-a_n=1$ 又由 $a_1=1$ 则 $a_n=a_1+(n-1)\times 1=n$

法二: 当 $n=1$ 时, $a_1^3=S_1^2$ 又由 $a_n>0$ 则 $a_1=1$; 当 $n=2$ 时, $a_1^3+a_2^3=S_2^2$ 则 $a_2=2$

当 $n=3$ 时, $a_1^3+a_2^3+a_3^3=S_3^2$ 则 $a_3=3$ 猜想 $a_n=n$ 为所求 假设 $n=k(k\in\mathbb{N}^*)$ 时成立,

则 $a_k=k$ 则 $S_k=\frac{k(k+1)}{2}$ 由 $a_1^3+\dots+a_k^3=S_k^2$ ①, $a_1^3+\dots+a_{k+1}^3=S_{k+1}^2$ ②, 则 ②-① 得

$a_{k+1}^3=S_{k+1}^2-S_k^2$ 则 $a_{k+1}^3=(S_k+a_{k+1})^2-S_k^2=a_{k+1}^2+2a_{k+1}S_k$ 由 $a_{k+1}>0$

则 $a_{k+1}^2=a_{k+1}+2S_k$ 则 $a_{k+1}^2-a_{k+1}-2S_k=0$ 则 $a_{k+1}=\frac{1\pm\sqrt{1^2-4\times(-2S_k)}}{2}=\frac{1\pm\sqrt{2k(k+1)^2}}{2}$

由 $a_{k+1}>0$ 则 $a_{k+1}=\frac{1+2k+1}{2}=k+1$ 则当 $n=k$ 成立时, $n=k+1$ 也成立.

(13) 由 $a_{n+1}a_n = 4S_n$ ①, 则 $a_{n+2}a_{n+1} = 4S_{n+1}$ ② 由 ② - ① 则 $(a_{n+2} - a_n)a_{n+1} = 4a_{n+1}$

由 $a_{n+1} > 0$ 则 $a_{n+2} - a_n = 4$. 当 $n=1$ 时, $a_1a_2 = 4S_1$ 则 $a_2 = 4$

当 n 为偶数时, $\begin{cases} a_{n+2} - a_n = 4 \\ a_4 - a_2 = 4 \end{cases}$ 则 $a_{n+2} - a_2 = 4 \times \frac{n}{2} = 2n$ 则 $a_{n+2} = 2n + 4$

当 $n \geq 4$ 时, $a_n = 2(n-2) + 4 = 2n$, 当 $n=2$ 时 $a_2 = 4$ 符合左式

当 n 为奇数时, $\begin{cases} a_{n+2} - a_n = 4 \\ a_3 - a_1 = 4 \end{cases}$ 则 $a_{n+2} - a_1 = 4 \times \frac{n+1}{2} = 2(n+1)$ 则 $a_{n+2} = 2n + 3$

当 $n \geq 3$ 时, $a_n = 2(n-2) + 3 = 2n - 1$, 当 $n=1$ 时, $a_1 = 1$ 符合左式.

则 $a_n = \begin{cases} 2n-1, n \text{ 为奇数} \\ 2n, n \text{ 为偶数} \end{cases}$

(14) 由 $a_{n+1} + S_{n+1}S_n = 0$ 则 $S_{n+1} - S_n + S_{n+1}S_n = 0$ 由 $a_n > 0$, 则 $S_n > 0$ 则 $\frac{1}{S_n} - \frac{1}{S_{n+1}} + 1 = 0$

则 $\frac{1}{S_{n+1}} - \frac{1}{S_n} = -1$ 则 $\frac{1}{S_n} = \frac{1}{S_1} + (n-1) \times (-1) = n$ 则 $S_n = \frac{1}{n}$ 当 $n \geq 2$ 时, $a_n = S_n - S_{n-1} = \frac{1}{n} - \frac{1}{n-1}$ 则 $a_n = \begin{cases} 1, n=1 \\ \frac{1}{n} - \frac{1}{n-1}, n \geq 2 \end{cases}$

(15) 由 $\frac{1}{a_1a_2} + \frac{1}{a_2a_3} + \dots + \frac{1}{a_na_{n+1}} = \frac{1}{6} - \frac{1}{3a_{n+1}}$ ①, 则 $\frac{1}{a_1a_2} + \dots + \frac{1}{a_{n+1}a_{n+2}} = \frac{1}{6} - \frac{1}{3a_{n+2}}$ ②

由 ② - ① 则 $\frac{1}{a_{n+1}a_{n+2}} = \frac{1}{3a_{n+1}} - \frac{1}{3a_{n+2}} = \frac{a_{n+2} - a_{n+1}}{3a_{n+1}a_{n+2}}$ 则 $a_{n+2} - a_{n+1} = 3$

由 $\frac{1}{a_1a_2} = \frac{1}{6} - \frac{1}{3a_2}$ ③ 又由 $a_1 = a_3$, $a_3 - a_2 = 3$ 则 $a_1 = a_2 + 3$ 则 由 ③, 得 $\frac{1}{a_2(a_2+3)} = \frac{1}{6} - \frac{1}{3a_2}$

则 $a_2^2 + a_2 - 12 = (a_2 - 3)(a_2 + 4) = 0$ 则 $a_2 = 3$ 或 -4 (舍) 由 $a_n = a_2 + (n-2) \times 3 = 3(n-1) (n \geq 2)$

当 $n=1$ 时, $a_1 = 6$ 不符合上式 综上 $a_n = \begin{cases} 3n-3, n \geq 2 \\ 6, n=1 \end{cases}$

(16) 由 $(1+\frac{1}{a_1})(1+\frac{1}{a_2}) \dots (1+\frac{1}{a_n}) = 2^{\frac{n(n+1)}{2}}$ ① 则 $(1+\frac{1}{a_1})(1+\frac{1}{a_2}) \dots (1+\frac{1}{a_{n+1}}) = 2^{\frac{(n+1)(n+2)}{2}}$ ②

由 ② ÷ ①, 则 $1 + \frac{1}{a_{n+1}} = 2^{\frac{(n+1)(n+2)}{2} - \frac{n(n+1)}{2}} = 2^{n+1}$ 则 $\frac{1}{a_{n+1}} = 2^{n+1} - 1$ 则 $a_{n+1} = \frac{1}{2^{n+1} - 1}$

当 $n \geq 2$ 时, 则 $a_n = \frac{1}{2^n - 1}$ 当 $n=1$ 时, 由 $(1+\frac{1}{a_1}) = 2^{\frac{1 \times 2}{2}} = 2$ 则 $a_1 = 1$ 符合左式

则 $a_n = \frac{1}{2^n - 1}$

例 4: (1) 当 $n=1$ 时, $a_1b_1 = 2(b_1-3)a_1 + 8$, 则 $b_1 = 2$; 当 $n=2$ 时, $a_1b_1 + a_2b_2 = 2(b_2-3)a_2 + 8$, 则 $a_2 = 4$

又由 $\{a_n\}$ 为等比数列, 则公比 $q = \frac{a_2}{a_1} = \frac{4}{1} = 4$ 则 $a_n = a_1 q^{n-1} = 2 \cdot 4^{n-1} = 2^n$

由 $b_1 \cdot 2^1 + b_2 \cdot 2^2 + \dots + b_n \cdot 2^n = 2^{n+1}(b_n-3) + 8$ ①, 则 $b_1 \cdot 2^1 + \dots + b_{n+1} \cdot 2^{n+1} = 2^{n+2}(b_{n+1}-3) + 8$ ②

又由 ② - ①, 则 $b_{n+1} \cdot 2^{n+1} = 2^{n+2}(b_{n+1}-3) - 2^{n+1}(b_n-3)$ 则 $b_{n+1} = b_n + 3$ 又由 $b_1 = 2$

则 $b_n = 2 + 3(n-1) = 3n - 1$

(2) 当 $n=1$ 时, $a_1b_1 = 4a_1 + b_1 - 3$, 则 $b_1 = 2$; 当 $n=2$ 时, $a_1b_1 + a_2b_2 = 4a_2 + b_2 - 3$, 则 $a_2 = \frac{1}{4}$

又由 $\{a_n\}$ 为等比数列, 则公比 $q = \frac{a_2}{a_1} = \frac{1/4}{1} = \frac{1}{4}$ 则 $a_n = a_1 q^{n-1} = \frac{1}{4} \cdot (\frac{1}{4})^{n-1} = (\frac{1}{4})^n$

由 $\frac{b_1}{2} + \frac{b_2}{2^2} + \dots + \frac{b_n}{2^n} = (\frac{1}{2})^{n-2} + b_n - 3$ 同乘 2^n 则 $b_1 + 2b_2 + \dots + 2^{n-1}b_n = (b_n-3) \cdot 2^n + 4$ ①

则 $b_1 + 2b_2 + \dots + 2^n b_{n+1} = (b_{n+1}-3) \cdot 2^{n+1} + 4$ ② 由 ② - ① 则 $2^n b_{n+1} = (b_{n+1}-3) \cdot 2^{n+1} - (b_n-3) \cdot 2^n$

则 $b_{n+1} = 2(b_{n+1}-3) - (b_n-3)$ 则 $b_{n+1} - b_n = 3$ 又由 $b_1 = 2$ 则 $b_n = b_1 + 3(n-1) =$

$3n - 1$ 则 $b_n = 3n - 1$

例5: 由 $a_{n+1} + b_n = 2n$ 则 $a_{n+2} + b_{n+1} = 2(n+1)$ ① 又由 $a_n + b_{n+1} = 2^n$ ② 由 ①-②得,

$$a_{n+2} - a_n = 2(n+1) - 2^n \quad \text{由 } a_2 + b_1 = 2 \times 1, a_1 + b_2 = 2^1, a_1 = b_1 = 0 \quad \text{则 } a_2 = b_2 = 2$$

当 n 为偶数时, 由 $\begin{cases} a_{n+2} - a_n = 2(n+1) - 2^n \\ \dots \\ a_4 - a_2 = 2 \times (2+1) - 2^2 \end{cases}$ 则 $a_{n+2} - a_2 = 2 \times \frac{(3+n+1) \frac{n}{2}}{2} - \frac{2^2[(\frac{n}{2})^2 - 1]}{2^2 - 1}$

$$= \frac{n(n+4)}{2} - \frac{4}{3}(2^n - 1)$$

$$\text{则 } a_{n+2} = \frac{n(n+4)}{2} - \frac{4}{3}(2^n - 1) + 2 = \frac{n(n+4)}{2} - \frac{1}{3} \times 2^{n+2} + \frac{10}{3}$$

当 $n \geq 4$ 且 n 为偶数时, $a_n = \frac{(n-2)(n+2)}{2} - \frac{1}{3} \times 2^n + \frac{10}{3}$ 当 $n=2$ 时, $a_2 = 2$ 符合左式

$$\text{则 } a_n = \frac{n^2}{2} - \frac{1}{3} \times 2^n + \frac{4}{3} \quad (n \text{ 为偶数})$$

当 n 为奇数时, 由 $\begin{cases} a_{n+2} - a_n = 2(n+1) - 2^n \\ \dots \\ a_3 - a_1 = 2 \times (1+1) - 2^1 \end{cases}$ 则 $a_{n+2} - a_1 = 2 \times \frac{(2+n+1) \frac{n+1}{2}}{2} - \frac{2^1[(\frac{n+1}{2})^2 - 1]}{2^2 - 1}$

$$= \frac{(n+1)(n+3)}{2} - \frac{2}{3}(2^{n+1} - 1)$$

$$\text{则 } a_{n+2} = \frac{(n+1)(n+3)}{2} - \frac{1}{3} \times 2^{n+2} + \frac{2}{3}$$

当 $n \geq 3$ 且 n 为奇数时, $a_n = \frac{(n-1)(n+1)}{2} - \frac{1}{3} \times 2^n + \frac{1}{3} = \frac{n^2}{2} - \frac{1}{3} \times 2^n + \frac{1}{6}$ 当 $n=1$ 时, $a_1 = 0$ 符合左式

$$\text{则 } a_n = \frac{n^2}{2} - \frac{1}{3} \times 2^n + \frac{1}{6} \quad (n \text{ 为奇数})$$

当 n 为奇数时, 则 $n+1$ 为偶数, 由 $b_n = 2n - a_{n+1} = 2n - [\frac{(n+1)^2}{2} - \frac{1}{3} \times 2^{n+1} + \frac{4}{3}] = -\frac{(n-1)^2}{2} + \frac{1}{3} \times 2^{n+1} - \frac{4}{3} \quad (n \text{ 为奇数})$

当 n 为偶数时, 则 $n+1$ 为奇数, 由 $b_n = 2n - a_{n+1} = 2n - [\frac{(n+1)^2}{2} - \frac{1}{3} \times 2^{n+1} + \frac{1}{6}] = -\frac{(n-1)^2}{2} + \frac{1}{3} \times 2^{n+1} - \frac{1}{6}$

综上所述: $a_n = \begin{cases} \frac{n^2}{2} - \frac{2^n}{3} + \frac{1}{6}, & n \text{ 为奇数} \\ \frac{n^2}{2} - \frac{2^n}{3} + \frac{4}{3}, & n \text{ 为偶数} \end{cases}, b_n = \begin{cases} -\frac{(n-1)^2}{2} + \frac{2^{n+1}}{3} - \frac{4}{3}, & n \text{ 为奇数} \\ -\frac{(n-1)^2}{2} + \frac{2^{n+1}}{3} - \frac{1}{6}, & n \text{ 为偶数} \end{cases}$

例6: (1) 当 $n=1$ 时, $\frac{a_1 - b_1}{1} = 2$ 则 $b_1 = 1$; 当 $n=2$ 时, $\frac{a_2 - b_2}{2} = 2$, 则 $a_2 = 6$

又由 $\{a_n\}$ 是等差数列, 公差 $d = a_2 - a_1 = 6 - 3 = 3$. 则 $a_n = a_1 + 3(n-1) = 3n$

由 $a_n - b_n = 2n$ 则 $b_n = a_n - 2n = 3n - 2n = n$

(2) 设 $\{a_n\}$ 公差为 d ($d \neq 0$) 则 $a_3 = a_1 + 2d = 4$ 由 $a_4 = S_3$ 则 $a_1 + 3d = 3a_1 + \frac{3 \times 2}{2}d$ ②

由 ①② 联立, 则 $\begin{cases} a_1 = 0 \\ d = 2 \end{cases}$ 则 $a_n = a_1 + 2(n-1) = 2(n-1)$ 则 $S_n = \frac{(0+2n-2)n}{2} = n(n-1)$

由 $(S_{n+1} + b_n)^2 = (S_{n+2} + b_n)(S_n + b_n)$ 则 $S_{n+1}^2 + 2b_n S_{n+1} = S_{n+2} S_n + b_n(S_n + S_{n+2})$

$$\text{则 } b_n(2S_{n+1} - S_n - S_{n+2}) = S_{n+2} S_n - S_{n+1}^2 \quad \text{则 } b_n = \frac{S_{n+2} S_n - S_{n+1}^2}{2S_{n+1} - S_n - S_{n+2}} = \frac{S_{n+2} S_n - S_{n+1}^2}{S_{n+1} + S_n + S_{n+1} - S_{n+2}} = \frac{S_{n+2} S_n - S_{n+1}^2}{a_{n+1} - a_{n+2}}$$

$$= n(n+1) \quad \text{则 } a_n = 2n - 2, b_n = n(n+1)$$

(3) 由 $\{b_n\}$ 为等差数列, 则 $b_2 = b_1 + b_3$ 则 $\frac{S_2}{a_2} = \frac{S_1}{a_1} + \frac{S_3}{a_3}$ 设 $\{a_n\}$ 公差为 d .

则 $\frac{S_2}{a_2} = 1 + \frac{S_3}{a_3}$ 则 $a_3 S_2 = a_2 a_3 + a_2 S_3$ 则 $(a_1 + 2d)(2a_1 + \frac{3 \times 2}{2}d) = (a_1 + d)(a_1 + 2d) + (a_1 + d)(3a_1 + \frac{3 \times 2}{2}d)$

解得 $d = a_1 = 1$ 则 $a_n = a_1 + 1 \times (n-1) = n$, $S_n = \frac{n(n+1)}{2}$ 则 $b_n = \frac{S_n}{a_n} = \frac{\frac{n(n+1)}{2}}{n} = \frac{n+1}{2}$

例7: (1) 当 $n=1$ 时, $a_1 = S_1 = T_1$, 则 $\frac{2}{a_1} + \frac{1}{a_1} = 2$ 则 $a_1 = \frac{3}{2}$ 当 $n \geq 2$ 时, $S_n = \frac{T_n}{T_{n-1}}$

$$\text{则 } \frac{2T_{n-1}}{T_n} + \frac{1}{T_n} = 2 \quad \text{则 } 2T_n = 2T_{n-1} + 1, \text{ 则 } T_n = T_{n-1} + \frac{1}{2} \text{ (由 } T_1 = \frac{3}{2}, \text{ 则 } T_n = \frac{3}{2} + \frac{1}{2}(n-1) = \frac{n+2}{2}$$

$$\text{当 } n \geq 2 \text{ 时, } S_n = \frac{T_n}{T_{n-1}} = \frac{\frac{n+2}{2}}{\frac{n+1}{2}} = \frac{n+2}{n+1} \quad \text{当 } n=1 \text{ 时, } S_1 = \frac{3}{2} \text{ 符合左式} \quad \text{则 } S_n = \frac{n+2}{n+1} = 1 + \frac{1}{n+1}$$

$$\text{当 } n \geq 2 \text{ 时, } a_n = S_n - S_{n-1} = (1 + \frac{1}{n+1}) - (1 + \frac{1}{n}) = \frac{1}{n+1} - \frac{1}{n} \quad \text{当 } n=1 \text{ 时, } a_1 = \frac{3}{2} \text{ 不符合左式}$$

$$\text{则 } a_n = \begin{cases} \frac{3}{2}, & n=1 \\ \frac{1}{n+1} - \frac{1}{n}, & n \geq 2 \end{cases}$$

$$(2) \text{ 当 } n=1 \text{ 时, } a_1 = S_1 = T_1, \text{ 则 } \frac{2}{a_1} + \frac{1}{a_1} = 1 \quad \text{则 } a_1 = 3 \quad \text{当 } n \geq 2 \text{ 时, } S_n = \frac{T_n}{T_{n-1}}$$

$$\text{则 } \frac{2T_{n-1}}{T_n} + \frac{1}{T_n} = 1 \quad \text{则 } T_n = 2T_{n-1} + 1 \text{ (} n \geq 2 \text{)} \quad \text{则 } T_n + 1 = 2(T_{n-1} + 1) \quad \text{又由 } T_1 + 1 = 4 \neq 0$$

$$\text{则 } T_n + 1 = 2^{n-1}(T_1 + 1) = 2^{n-1} \quad \text{则 } T_n = 2^{n-1} - 1 \quad \text{当 } n \geq 2 \text{ 时, } S_n = \frac{T_n}{T_{n-1}} = \frac{2^{n-1} - 1}{2^{n-2} - 1} = \frac{2(2^{n-1}) + 1}{2^{n-1} - 1}$$

$$= 2 + \frac{1}{2^{n-1} - 1} \quad \text{当 } n=1 \text{ 时, } S_1 = 3 \text{ 符合左式} \quad \text{当 } n \geq 2 \text{ 时, } a_n = S_n - S_{n-1} = (2 + \frac{1}{2^{n-1} - 1}) - (2 + \frac{1}{2^{n-2} - 1})$$

$$= \frac{1}{2^{n-1} - 1} - \frac{1}{2^{n-2} - 1} \quad \text{当 } n=1 \text{ 时, } a_1 = 3 \text{ 不符合左式} \quad \text{则 } a_n = \begin{cases} 3, & n=1 \\ \frac{1}{2^{n-1} - 1} - \frac{1}{2^{n-2} - 1}, & n \geq 2 \end{cases}$$

$$(3) \text{ 由 } T_n = S_n \cdot A_{2n}^n \quad \text{由 } S_n = \frac{T_n}{T_{n-1}} \text{ (} n \geq 2 \text{)} \quad \text{则 } T_n = \frac{T_n}{T_{n-1}} A_{2n}^n \quad \text{则 } T_{n-1} = A_{2n}^n \text{ (} n \geq 2 \text{)}$$

$$\text{则 } T_n = A_{2(n+1)}^{n+1} = A_{2n+2}^{n+1} \quad \text{则 } S_n = \frac{T_n}{T_{n-1}} = \frac{A_{2n+2}^{n+1}}{A_{2n}^n} = \frac{(2n+2)!}{(n+1)!} \cdot \frac{n!}{(2n)!} = 4n+2 \text{ (} n \geq 2 \text{)}$$

$$\text{当 } n \geq 2 \text{ 时, } T_2 = S_2 A_4^2 = 12 S_2 \quad \text{又由 } T_2 = S_2 S_2 \quad \text{则 } S_1 = 12 \quad \text{由 } S_1 = 12 \text{ 不符合上式, 则 } S_n = \begin{cases} 12, & n=1 \\ 4n+2, & n \geq 2 \end{cases}$$

$$\text{当 } n \geq 3 \text{ 时, } a_n = S_n - S_{n-1} = 4n+2 - [4(n-1)+2] = 4 \quad \text{当 } n=2 \text{ 时, } S_2 = 4 \times 2 + 2 = 10 = a_1 + a_2 \quad \text{则 } a_2 = -2$$

$$\text{由 } a_1 = 12, a_2 = -2 \text{ 不符合上式} \quad \text{则 } a_n = \begin{cases} 12, & n=1 \\ -2, & n=2 \\ 4, & n \geq 3 \end{cases}$$

$$(4) \text{ 由 } T_n = S_n \cdot 2^{n^2+n} = \frac{T_n}{T_{n-1}} 2^{n^2+n} \quad \text{则 } T_{n-1} = 2^{n^2+n} \text{ (} n \geq 2 \text{)} \quad \text{则 } T_n = 2^{(n+1)(n+2)} \quad \text{由 } S_n = \frac{T_n}{T_{n-1}} = \frac{2^{(n+1)(n+2)}}{2^{n^2+n}} = 2^{2n+2}$$

$$= 2^{2(n+1)} \text{ (} n \geq 2 \text{)} \quad \text{由 } T_2 = S_1 S_2 = 2^6 S_2 \quad \text{则 } S_1 = 2^6 \quad \text{由 } S_1 \text{ 不符合左式, 则 } S_n = \begin{cases} 2^{2n+2}, & n \geq 2 \\ 2^6, & n=1 \end{cases}$$

$$\text{由 } a_1 = S_1 = 2^6, a_1 + a_2 = S_2 = 2^6 \quad \text{则 } a_2 = 0 \quad \text{当 } n \geq 3 \text{ 时, } a_n = S_n - S_{n-1} = 2^{2n+2} - 2^{2n} = 3 \times 2^{2n}$$

$$\text{综上 } a_n = \begin{cases} 2^6, & n=1 \\ 0, & n=2 \\ 3 \times 2^{2n}, & n \geq 3 \end{cases}$$

$$(5) \text{ 由 } T_n = S_n \cdot C_{2n}^n \quad \text{则 } T_n = \frac{T_n}{T_{n-1}} C_{2n}^n \quad \text{则 } T_{n-1} = C_{2n}^n \text{ (} n \geq 2 \text{)} \quad \text{则 } T_n = C_{2(n+1)}^{n+1} \quad \text{则 } S_n = \frac{T_n}{T_{n-1}} = \frac{C_{2n+2}^{n+1}}{C_{2n}^n}$$

$$= \frac{(2n+2)!}{(n+1)! (n+1)!} = \frac{4n+2}{n+1} = \frac{4(n+1)-2}{n+1} = 4 - \frac{2}{n+1} \text{ (} n \geq 2 \text{)} \quad \text{由 } T_2 = S_1 S_2 = S_2 C_4^2 = \quad \text{则 } S_1 = C_4^2 = 6$$

$$\text{则 } S_n = \begin{cases} 6, & n=1 \\ 4 - \frac{2}{n+1}, & n \geq 2 \end{cases} \quad \text{当 } n=2 \text{ 时, } S_2 = 4 - \frac{2}{3} = \frac{10}{3} = a_1 + a_2, \text{ 则 } a_2 = -\frac{8}{3} \quad \text{当 } n \geq 3 \text{ 时,}$$

$$a_n = S_n - S_{n-1} = 4 - \frac{2}{n+1} - (4 - \frac{2}{n}) = \frac{2}{n} - \frac{2}{n+1} = \frac{2}{n(n+1)} \quad \text{则 } a_n = \begin{cases} \frac{6}{3}, & n=1 \\ -\frac{8}{3}, & n=2 \\ \frac{2}{n(n+1)}, & n \geq 3 \end{cases}$$

例8: 若选①②, 则由 $\{\sqrt{S_n}\}$ 为等差数列, 则 $2\sqrt{S_2} = \sqrt{S_1} + \sqrt{S_3}$ 设 $\{a_n\}$ 公差为 d .

$$\text{则 } 2\sqrt{2a_1+d} = \sqrt{a_1} + \sqrt{3a_1+3d} \quad \text{则 } 4(2a_1+d) = a_1 + 3a_1 + 3d + 2\sqrt{a_1(3a_1+3d)} \quad \text{则 } 4a_1 + d = 2\sqrt{3a_1(a_1+d)}$$

$$\text{则 } (4a_1+d)^2 = 4 \times 3a_1(a_1+d) = 12a_1(a_1+d) \quad \text{则 } 4a_1^2 - 4a_1d + d^2 = (2a_1-d)^2 = 0 \quad \text{则 } d = 2a_1$$

$$\text{则 } a_2 = a_1 + d = 3a_1 \quad \text{证毕.}$$

若选①③, 则由 $a_2 = a_1 + d, a_2 = 3a_1$, 则 $d = 2a_1$, 则 $a_n = a_1 + (n-1)d =$

$$a_1 + (n-1) \times 2a_1 = (2n-1)a_1 \quad \text{则 } S_n = \frac{[a_1 + (2n-1)a_1]n}{2} = n^2 a_1 \quad \text{则 } \sqrt{S_n} = \sqrt{a_1} \cdot n \quad \text{则 } \sqrt{S_{n+1}} - \sqrt{S_n} =$$

$$\sqrt{a_1}(n+1) - \sqrt{a_1} \cdot n = \sqrt{a_1} \quad \text{则 } \{\sqrt{S_n}\} \text{ 为等差数列, 证毕}$$

若选②④，则由 $\{\sqrt{s_n}\}$ 为等差数列。则 $\sqrt{s_2} - \sqrt{s_1} = \sqrt{a_1 + a_2} - \sqrt{a_1} = \sqrt{4a_1} - \sqrt{a_1} = \sqrt{a_1}$

则 $\{\sqrt{s_n}\}$ 是以 $\sqrt{a_1}$ 为首项， $\sqrt{a_1}$ 为公差的等差数列 则 $\sqrt{s_n} = \sqrt{a_1} + (n-1)\sqrt{a_1} = n\sqrt{a_1}$ 则 $S_n = a_1 \cdot n^2$

当 $n \geq 2$ 时， $a_n = S_n - S_{n-1} = n^2 a_1 - (n-1)^2 a_1 = (2n-1)a_1$ ，当 $n=1$ 时， $a_1 = a_1$ 符合左式 则 $a_n = (2n-1)a_1$

由 $a_{n+1} - a_n = 2a_1$ 则 $\{a_n\}$ 为等差数列 证毕

例 9: (1) 由 $a_n = \frac{1}{n^2}$ 当 $n \geq 2$ 时，有 $a_n = \frac{1}{n^2} < \frac{1}{n(n-1)} = \frac{1}{n-1} - \frac{1}{n}$ 则 $S_1 = a_1 = \frac{1}{4} < \frac{7}{4}$, $S_2 = a_1 + a_2 = \frac{5}{4} < \frac{7}{4}$

$$S_n = a_1 + a_2 + \dots + a_n = \frac{1}{4} + \frac{1}{9} + \dots + \frac{1}{n^2} < \frac{1}{4} + \frac{1}{9} + (\frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \dots - \frac{1}{n-1} + \frac{1}{n}) = \frac{7}{4} - \frac{1}{n} < \frac{7}{4}$$

(2) 由 $a_n = \frac{1}{\sqrt{n}}$ 易知，有 $a_n = \frac{1}{\sqrt{n}} = \frac{2}{2\sqrt{n}}$ 由 $\sqrt{n+1} + \sqrt{n} < 2\sqrt{n} < \sqrt{n} + \sqrt{n+1}$ 则 $\frac{2}{\sqrt{n} + \sqrt{n+1}} < a_n < \frac{2}{\sqrt{n} + \sqrt{n-1}}$

则 $2(\sqrt{n+1} - \sqrt{n}) < a_n < 2(\sqrt{n} - \sqrt{n-1})$ 则 $2(\sqrt{n+1} - \sqrt{n} + \dots - \sqrt{2} + \sqrt{1}) < S_n < 2(\sqrt{n} - \sqrt{n-1} + \dots - \sqrt{1} + \sqrt{0})$

则 $2\sqrt{n+1} - 2 < S_n < 2\sqrt{n}$ (注: $\frac{1}{\sqrt{n}}$ 无法直接求和，要放缩成例 2(5) 形式再求和)

(3) 由 $a_n = \frac{1}{n\sqrt{n}} = \frac{2}{2n\sqrt{n}}$ 证: $2n\sqrt{n} > n\sqrt{n-1} + (n-1)\sqrt{n}$ ($n \geq 2$) 则证: $(n+1)\sqrt{n} > n\sqrt{n-1}$

则证: $n+1 > \sqrt{n(n-1)}$ 则证: $(n+1)^2 > n(n-1)$ 则证: $3n+1 > 0$ 易知 $n \geq 2$ 时， $3n+1 > 0$ 成立。

则 $a_n = \frac{2}{2n\sqrt{n}} < \frac{2}{n\sqrt{n-1} + (n-1)\sqrt{n}} = \frac{2}{\sqrt{n(n-1)}(\sqrt{n} + \sqrt{n-1})} = \frac{2(\sqrt{n} - \sqrt{n-1})}{\sqrt{n(n-1)}} = 2(\frac{1}{\sqrt{n-1}} - \frac{1}{\sqrt{n}})$ ($n \geq 2$)

由 $n=1$ 时， $S_1 = \frac{1}{1\sqrt{1}} = 1 < 3$ 则 $n \geq 2$ 时， $S_n = a_1 + a_2 + \dots + a_n = 1 + 2(\frac{1}{\sqrt{2-1}} - \frac{1}{\sqrt{2}} + \dots - \frac{1}{\sqrt{n-1}} + \frac{1}{\sqrt{n}}) = 1 + 2(1 - \frac{1}{\sqrt{n}}) = 3 - \frac{2}{\sqrt{n}} < 3$

(4) 由 $a_n = \frac{2^n}{(2^n-1)^2} = \frac{2^n}{(2^n-1)(2^n-1)} = \frac{2^n}{(2^n-1)(2^{n-1}-\frac{1}{2})} = \frac{2^{n+1}}{(2^n-1)(2^{n-1}-\frac{1}{2})} < \frac{2^{n+1}}{2^{n-1}(2^{n-1}-1)} = \frac{1}{2^{n-1}-1} - \frac{1}{2^{n-1}} (n \geq 2)$

则 $n=1$ 时， $S_1 = \frac{2^1}{(2^1-1)^2} = 2 < 3$ 当 $n \geq 2$ 时， $S_n = 2 + (\frac{1}{2^{n-1}-1} - \frac{1}{2^{n-1}} + \dots - \frac{1}{2^{n-1}} + \frac{1}{2^{n-1}}) = 3 - \frac{1}{2^{n-1}} < 3$

(5) 由 $a_n = \frac{2}{3^n-1}$ 设 $\frac{2}{3^n-1} \leq p \cdot (\frac{1}{3})^n$ 成立，则 $S_n \leq p(\frac{1}{3})^1 + p(\frac{1}{3})^2 + \dots + p(\frac{1}{3})^n = p \frac{(\frac{1}{3})^1 (1 - (\frac{1}{3})^n)}{1 - \frac{1}{3}} = \frac{p}{2} (1 - (\frac{1}{3})^n) < \frac{p}{2}$ 令 $\frac{p}{2} = \frac{3}{2}$ 则 $p=3$ 下证: 当 $p=3$ 时， $\frac{2}{3^n-1} \leq p \cdot (\frac{1}{3})^n$ 成立。则证: $\frac{2}{3^n-1} \leq (\frac{1}{3})^{n-1}$ 成立

则证: $2 \times 3^{n-1} \leq 3^n - 1$ 则证: $1 \leq 3^{n-1}$ 成立。易知， $n \in \mathbb{N}^*$ 时， $3^{n-1} \geq 1$ 成立。则命题得证。

(6) 设 T_n 是 $\{b_n\}$ 的前 n 项和，令 $T_n = \frac{6^n-1}{5}$ 则 $n=1$ 时， $a_1 = T_1 = \frac{6^1-1}{5} = 1$

当 $n \geq 2$ 时， $a_n = T_n - T_{n-1} = \frac{6^n-1}{5} - \frac{6^{n-1}-1}{5} = \frac{6^n-6^{n-1}}{5} = \frac{6^{n-1}(6-1)}{5} = \frac{6^{n-1}}{5}$ 要证: $S_n \leq \frac{6^n-1}{5}$ 则证: $a_n \leq b_n$

则证: $\frac{6^n}{(n+1)(2n+1)} \leq \frac{6^{n-1}}{5}$ 则证: $6 \leq (n+1)(2n+1)$ 成立 易知: $(n+1)(2n+1) \geq 6$ 成立

(7) 由 $a_n = \ln(1 + \frac{1}{n^2})$ 设 $f(x) = \ln(1+x) - x$ ，则 $f'(x) = \frac{1}{x+1} - 1 = \frac{-x}{x+1}$ 当 $x > 0$ 时， $f'(x) < 0$

则 $f(x)$ 单调递减。又由 $f(x) < f(0) = 0$ 则 $\ln(1+x) - x < 0$ 在 $x > 0$ 恒成立 则 $\ln(1+x) < x$ 恒成立

令 $x = \frac{1}{n^2}$ ($n \in \mathbb{N}^*$) 则 $\ln(1 + \frac{1}{n^2}) < \frac{1}{n^2}$ 又由 $\frac{1}{n^2} < \frac{1}{n^2-1} = \frac{4}{4n^2-4} = 2(\frac{1}{2n-1} - \frac{1}{2n+1})$

则 $S_n = a_1 + \dots + a_n < \frac{1}{4} + \dots + \frac{1}{n^2} < 2(\frac{1}{1} - \frac{1}{3} + \dots - \frac{1}{2n-1} + \frac{1}{2n+1}) = 2 - \frac{2}{2n+1} < 2$

(8) 设 T_n 是 $\{b_n\}$ 前 n 项和，令 $T_n = \frac{n(n+1)}{4}$ 则 $n=1$ 时， $a_1 = T_1 = 0$ 当 $n \geq 2$ 时， $b_n =$

$T_n - T_{n-1} = \frac{n(n+1)}{4} - \frac{(n-1)n}{4} = \frac{n}{2}$ 要证: $S_n \leq \frac{n(n+1)}{4}$ 则证: $a_n \leq b_n$ 则证: $\frac{\ln(n)}{n+1} \leq \frac{n-1}{2}$

则证: $\ln(n) \leq \frac{n^2-1}{2}$ 则证: $2\ln(n) \leq n^2-1$ 则证: $\ln(n^2) \leq n^2-1$ 设 $f(x) = x^2 - 1 - \ln x$

则 $f(x) = 1 - \frac{1}{x} = \frac{x-1}{x}$ 当 $x > 1$ 时, $f(x) > 0$ 则 $f(x)$ 单调递增. 由 $f(x) > f(1) = 0$

则 $x - 1 - \ln x > 0$ 在 $x > 1$ 时恒成立. 则 $\ln x < x - 1$ 令 $x = n^2 (n \in \mathbb{N}^*)$ 则 $\ln n^2 < n^2 - 1$ 得证.

(9) 设 T_n 是 $\{b_n\}$ 的前 n 项和. 令 $T_n = \ln(n+1)$ 则 $b_1 = T_1 = \ln 2$ 当 $n \geq 2$ 时,

$b_n = T_n - T_{n-1} = \ln(n+1) - \ln n = \ln\left(\frac{n+1}{n}\right) = \ln\left(1 + \frac{1}{n}\right)$ 要证: $S_n > \ln(n+1)$, 则证: $a_n > b_n$

则证: $\frac{1}{\sqrt{n^2+n}} > \ln\left(1 + \frac{1}{n}\right)$ 由 $\frac{1}{\sqrt{n^2+n}} = \frac{1}{n} \cdot \frac{1}{\sqrt{1+\frac{1}{n}}} = \frac{\frac{1}{n}+1}{\sqrt{1+\frac{1}{n}}} = \sqrt{1+\frac{1}{n}} - \frac{1}{\sqrt{1+\frac{1}{n}}}$ 则证: $\sqrt{1+\frac{1}{n}} - \frac{1}{\sqrt{1+\frac{1}{n}}} > \ln\left(1 + \frac{1}{n}\right)$

设: $f(x) = \sqrt{x} - \frac{1}{\sqrt{x}} - \ln x$ 则 $f'(x) = \frac{1}{2\sqrt{x}} + \frac{1}{2x\sqrt{x}} - \frac{1}{x} = \frac{x-2\sqrt{x}+1}{2x\sqrt{x}} = \frac{(\sqrt{x}-1)^2}{2x\sqrt{x}} > 0$ 则 $f(x)$ 单调递增.

又由 $f(x) > f(1) = 0$. 则 $x > 1$ 时, $\sqrt{x} - \frac{1}{\sqrt{x}} - \ln x > 0$ 恒成立. 令 $x = 1 + \frac{1}{n} (n \in \mathbb{N}^*)$

则 $\sqrt{1+\frac{1}{n}} - \frac{1}{\sqrt{1+\frac{1}{n}}} > \ln\left(1 + \frac{1}{n}\right)$ 得证

(10) 要证: $\sum_{i=1}^n \frac{1}{i+1} < S_n < \sum_{i=1}^n \frac{1}{i}$ 则证: $\frac{1}{n+1} < \ln\left(1 + \frac{1}{n}\right) < \frac{1}{n}$ 由 $\frac{1}{n+1} = \frac{1}{1+\frac{1}{n}}$

设 $f(x) = \ln(1+x) - \frac{x}{1+x}$, $g(x) = x - \ln(1+x)$. 则 $f'(x) = \frac{1}{x+1} - \frac{1+x-x}{(x+1)^2} = \frac{1}{x+1} - \frac{1}{(x+1)^2} = \frac{x}{(x+1)^2}$ 当 $x > 0$ 时, $f'(x) > 0$. 则 $f(x)$ 单调递增. 由 $f(x) > f(0) = 0$

则 $x > 0$ 时, $\ln(1+x) - \frac{x}{1+x} > 0$ 恒成立. 令 $x = \frac{1}{n} (n \in \mathbb{N}^*)$ 则 $\ln\left(1 + \frac{1}{n}\right) > \frac{\frac{1}{n}}{1+\frac{1}{n}} = \frac{1}{n+1}$ 成立.

又由 $g'(x) = 1 - \frac{1}{x+1} = \frac{x}{x+1}$ 当 $x > 0$ 时, $g'(x) > 0$ 则 $g(x)$ 单调递增. 又由 $g(x) > g(0) = 0$

则 $x > 0$ 时, $x - \ln(1+x) > 0$ 恒成立. 令 $x = \frac{1}{n} (n \in \mathbb{N}^*)$ 则 $\ln\left(1 + \frac{1}{n}\right) < \frac{1}{n}$ 成立.

则 $\frac{1}{n+1} < \ln\left(1 + \frac{1}{n}\right) < \frac{1}{n}$ 得证.

例 10: (1) 假设存在 $a_m, a_p, a_q (m, p, q \in \mathbb{N}^* \text{ 且 } m \neq p \neq q)$ 三项构成等差数列.

则 $2a_p = a_q + a_m$ 又由 $a_m = a_1 \cdot 2^{m-1}$, $a_p = a_1 \cdot 2^{p-1}$, $a_q = a_1 \cdot 2^{q-1}$ 则 $2 \cdot 2^{p-1} = 2^{m-1} + 2^{q-1}$

则 $2^p = 2^{m-1} + 2^{q-1}$ 则 $2^{p-2+1} = 2^{m-2} + 1$ 易知 2^{p-2+1} , 2^{m-2} 均为偶数, 则该式不成立.

则 $\{a_n\}$ 中任意三项均不构成等差数列

(2) 假设 $\{a_n+1\}$ 是等比数列. 则 a_1+1, a_2+1, a_3+1 构成等比数列

则 $(a_2+1)^2 = (a_1+1)(a_3+1)$ 则 $a_2^2 + 2a_2 = a_1a_3 + a_1 + a_3$ 由 $\{a_n\}$ 为等比数列, 则 $a_2^2 = a_1a_3$

则 $2a_2 = a_1 + a_3$ 则 $2a_1 \cdot 2 = a_1 + a_1 \cdot 2^2$ 则 $4a_1 = 5a_1$ 则 $a_1 = 0$ 由 $\{a_n\}$ 为等比数列,

则 $a_n \neq 0$. 则 $\{a_n+1\}$ 不是等比数列

(3) 假设 $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ 是等差数列 则 $\frac{2}{b} = \frac{1}{a} + \frac{1}{c} = \frac{a+c}{ac}$ 又由 a, b, c 是等差数列

则 $2b = a+c$ 则 $\frac{2}{\frac{a+c}{2}} = \frac{a+c}{ac}$ 则 $(a+c)^2 = 4ac$ 则 $(a-c)^2 = 0$ 则 $a=c$, 与 $\{a_n\}$ 是

公差不为 0 的等差数列矛盾 则 $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ 不是等差数列

(4) 假设 $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ 是等比数列. 则 $\frac{1}{b^2} = \frac{1}{a} \cdot \frac{1}{c}$ 则 $b^2 = ac$ 又由 a, b, c

是等差数列 则 $\left(\frac{a+c}{2}\right)^2 = ac$ 则 $(a-c)^2 = 0$ 则 $a=c$, 与 $\{a_n\}$ 是公差不为 0

的等差数列矛盾 则 $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ 不是等比数列

例11: (1) 设 $\{a_n\}$ 的第 p 项 ($p \in \mathbb{N}^*$) 与 $\{b_n\}$ 的第 q 项 ($q \in \mathbb{N}^*$) 相等 则 $a_p = b_q$

则 $3p-1 = 2q+1$ 则 $q = \frac{3p-2}{2}$ 由 $q \in \mathbb{N}^*$, 则 $3p$ 必为偶数, 则 p 为偶数 则 $C_n = a_{2n} = 3 \times (2n) - 1 = 6n - 1$

(2) 设 $\{a_n\}$ 的第 p 项 ($p \in \mathbb{N}^*$) 与 $\{b_n\}$ 的第 q 项 ($q \in \mathbb{N}^*$) 相等, 则 $a_p = b_q$ 则 $3p-1 = 2^q$

则 $p = \frac{2^q+1}{3} = \frac{(3-1)^q+1}{3} = \frac{1}{3} \left[\sum_{i=1}^q C_q^i 3^i (-1)^{q-i} + (-1)^q + 1 \right] = \sum_{i=1}^q C_q^i 3^{i-1} (-1)^{q-i} + \frac{(-1)^q+1}{3}$ 由 $p \in \mathbb{N}^*$

则 $\frac{(-1)^q+1}{3}$ 应为整数 则 q 为奇数 则 $C_n = b_{2n-1} = 2^{2n-1}$

(3) 设 $\{a_n\}$ 的第 p 项 ($p \in \mathbb{N}^*$) 与 $\{b_n\}$ 的第 q 项 ($q \in \mathbb{N}^*$) 相等, 则 $a_p = b_q$, 则 $2p-1 = 2^q-1$

则 $p = 2^{q-1}$ 则 $\forall q \in \mathbb{N}^*$, 都 $\exists p \in \mathbb{N}^*$, 使 $p = 2^{q-1}$ 则 $\{b_n\}$ 的所有项都在 $\{a_n\}$ 中.

设 S_{20} 中去掉了 m ($m \in \mathbb{N}^*$) 项 $\{a_n\}$ 与 $\{b_n\}$ 的公共项. 则 $S_{20} = a_1 + a_2 + \dots + a_{20+m} - (b_1 + b_2 + \dots + b_m)$ 由 $b_m \leq a_{20+m} < b_{m+1}$ 则 $2^m - 1 \leq 2(20+m) - 1 < 2^{m+1} - 1$ 则 $2^m \leq m + 39 < 2^{m+1}$

则 $m = 5$. 则 $S_{20} = a_1 + \dots + a_{25} - (b_1 + b_2 + \dots + b_5) = \frac{(1+2 \times 25-1) \times 25}{2} - \left[\frac{2(2^5-1)}{2-1} - 5 \right] = 568$

(4) 设 $\{C_n\}$ 中在 a_{k+1} 之前 (不含 a_{k+1}) 所有项的项的个数记为 P_k

则 $P_k = 2 + 3 + \dots + k+1 = \frac{k(2+k+1)}{2} = \frac{k(k+3)}{2}$ 由 $P_m < 25 < P_{m+1}$ 则 $\frac{m(m+3)}{2} < 25 < \frac{(m+1)(m+4)}{2}$

则 $m = 5$ 则 $S_{25} = a_1 + a_2 + a_3 + a_4 + a_5 + a_6 + C_1 + 2C_2 + 3C_3 + 4C_4 + 5C_5 + 4C_6 = (a_1 + a_2 + \dots + a_6) +$

$(C_1 + 2C_2 + \dots + 5C_5) + 4C_6 = \frac{(1+2 \times 6-1) \times 6}{2} + [2^1 + 2 \times 2^2 + \dots + 5 \times 2^5 - (1+2+\dots+5)] + 4 \times (2^6-1) = 531$

(5) 设 $\{C_n\}$ 中 p 项是 $\{a_n\}$ 中的项, q 项是 $\{b_n\}$ 中的项 ($p, q \in \mathbb{N}^*$) 则 $p+q=27$

若 $C_{27} = 2^q$, 则 $C_{26} = 2^q - 1$ 则 $S_{27} = (2^1 + 2^2 + \dots + 2^q) + (1 + 3 + \dots + 2^q - 1)$ 由 $\{a_n\}$ 的第 p 项

为 $2^q - 1$ 则 $2p+1 = 2^q - 1$ 则 $p = \frac{2^q-2}{2} = 2^{q-1} - 1$ 又由 $p+q=27$ 则 $2^{q-1} - q - 28 = 0$.

又由 $q \in \mathbb{N}^*$, 则 q 不存在.

若 $C_{27} = 2p+1$, 则 $S_{27} = (1+3+\dots+2p+1) + 2^1 + \dots + 2^q$ 其中 $2^q < 2p+1 < 2^{q+1}$, 又由 $p+q=27$

则 $2^q < 55-2q < 2^{q+1}$ 又由 $q \in \mathbb{N}^*$, 则 $q=5$ 则 $S_{27} = \frac{(1+2p+1)p}{2} + \frac{2(2^q-1)}{2-1} = p(p+1) + 2^q - 2$

$= 22 \times 23 + 2^6 - 2 = 568$

(6) 由 $2^k \leq 3n-1 \leq 2^{k+1}$ 则 $\frac{2^k+1}{3} \leq n \leq \frac{2^{k+1}+1}{3}$

当 k 为奇数时, $\frac{2^k+1}{3}$ 为整数, 则 $2^{k+1}+1$ 除以 3 的余数为 2 则 $\frac{2^{k+1}+1}{3}$ 整数部分为 $\frac{2^{k+1}+1-2}{3} = \frac{2^{k+1}-1}{3}$ 由 $\frac{2^k+1}{3} \leq n \leq \frac{2^{k+1}-1}{3}$ 则 $C_k = \frac{2^{k+1}-1}{3} - \frac{2^k+1}{3} + 1 = \frac{2^k+1}{3}$ (k 为奇数)

当 k 为偶数时, $\frac{2^{k+1}+1}{3}$ 为整数, 由 2^k+1 除以 3 的余数为 2, 则大于 $\frac{2^k+1}{3}$ 的最小整数为 $\frac{2^k+1+1}{3} = \frac{2^k+2}{3}$ 则 $C_k = \frac{2^{k+1}+1}{3} - \frac{2^k+2}{3} + 1 = \frac{2^k+2}{3}$ (k 为偶数)

综上 $C_k = \begin{cases} \frac{2^k+1}{3}, & k \text{ 为奇数} \\ \frac{2^k+2}{3}, & k \text{ 为偶数} \end{cases}$

(7) 由 $b_k = \frac{a_{k+1} - a_k}{k+1} = \frac{2^{k+1} - 2^k}{k+1} = \frac{2^k}{k+1}$

例12: 由 $2^n = (3-1)^n = \sum_{i=0}^n C_n^i 3^i (-1)^{n-i} + (-1)^n$, 则 $\frac{2^n}{3}$ 的整数部分为 $\sum_{i=1}^n C_n^i 3^{i-1} (-1)^{n-i}$,
 小数部分由 $(-1)^n$ 决定, 当 n 为奇数, $\frac{2^n}{3} = [\frac{2^n}{3}] + \frac{2}{3}$; 当 n 为偶数, $\frac{2^n}{3} = [\frac{2^n}{3}] + \frac{1}{3}$

即 $a_n = \begin{cases} \frac{2^n}{3} - \frac{2}{3}, n \text{ 为奇数} \\ \frac{2^n}{3} - \frac{1}{3}, n \text{ 为偶数} \end{cases}$ 当 n 为偶数时, 则 $S_n = \frac{1}{3}(2^1 + 2^2 + 2^3 + \dots + 2^n) - \frac{2}{3} \cdot \frac{n}{2} = \frac{2^{n+1}-2}{3} - \frac{1}{3}n$

当 n 为奇数时, $S_n = S_{n+1} - a_{n+1} = \frac{2^{n+2}-2}{3} - \frac{1}{2}(n+1) - (\frac{2^{n+1}}{3} - \frac{1}{3}) = \frac{2^{n+1}-1}{3} - \frac{n+1}{2}$

综上, $S_n = \begin{cases} \frac{2^{n+1}-2}{3} - \frac{1}{2}n, n \text{ 为偶数} \\ \frac{2^{n+1}-1}{3} - \frac{n+1}{2}, n \text{ 为奇数} \end{cases}$

例13: (1) 当 $n \geq 2$ 时, $a_{2n+4} - a_{2n+2} = \frac{1}{1+a_{2n+3}} - \frac{1}{1+a_{2n+1}} = \frac{1}{1+\frac{1}{1+a_{2n+2}}} - \frac{1}{1+\frac{1}{1+a_{2n}}}$
 $= \frac{a_{2n+2}+1}{a_{2n+2}+2} - \frac{a_{2n}+1}{a_{2n}+2} = \frac{a_{2n+2}-a_{2n}}{(a_{2n+2}+2)(a_{2n}+2)}$ 由 $a_1=1, a_{n+1} = \frac{1}{1+a_n}$, 则 $a_2=\frac{2}{3}, a_3=\frac{3}{5}, a_4=\frac{5}{8}$

则 $a_4 - a_2 = -\frac{1}{24} < 0$, 则 $a_{2n+2} - a_{2n} < 0$, 即 $\{a_{2n}\}$ 单调递减

(2) 由 $a_1 = \frac{1}{2}, a_{n+1} = \frac{1}{1+a_n}$, 则易知 $a_n > 0$, 则 $a_{n+1} = \frac{1}{1+a_n} < \frac{1}{1+0} = 1$, 则 $a_n < 1$

则 $a_{n+1} = \frac{1}{1+a_n} > \frac{1}{1+1} = \frac{1}{2}$, 又由 $a_1 = \frac{1}{2}$, 则 $\frac{1}{2} \leq a_n < 1$

(3) 由 $a_{n+2} - a_{n+1} = \frac{1}{1+a_{n+1}} - \frac{1}{1+a_n} = \frac{a_n - a_{n+1}}{(1+a_{n+1})(1+a_n)} = \frac{a_n - a_{n+1}}{2+a_n}$ 又由 $a_n \geq \frac{1}{2}$

则 $\frac{a_n - a_{n+1}}{2+a_n} \leq \frac{a_n - a_{n+1}}{2+\frac{1}{2}} = \frac{2}{5}(a_n - a_{n+1})$ 则 $|a_{n+2} - a_{n+1}| \leq \frac{2}{5}|a_{n+1} - a_n|$

又由 $|a_2 - a_1| = \frac{1}{6} \neq 0$ 即 $|a_{n+1} - a_n| \leq \frac{1}{6} \times (\frac{2}{5})^{n-1}$

例14: (1) 由 $n=1$ 时, $2\sqrt{a_1} = a_1 + c$ 则 $a_1 - 2\sqrt{a_1} = -c$ 则 $(\sqrt{a_1} - 1)^2 = 1 - c$ 则 $1 - c \geq 0$
 则 $c \leq 1$

(2) 法一: 由 $2\sqrt{S_n} = a_n$ 当 $n=1$ 时, $2\sqrt{a_1} = a_1$ 则 $a_1 = 4$ 或 0 (舍) 则 $S_1 = 4 = (1+1)^2$

当 $n \geq 2$ 时, 由 $2\sqrt{S_n} = a_n$ 则 $S_{n-1} = S_n - 2\sqrt{S_n}$ 则 $S_{n-1} + 1 = (\sqrt{S_n} - 1)^2$ 则 $\sqrt{S_{n-1} + 1} = \sqrt{S_n} - 1$

则 $\sqrt{S_n} = 1 + \sqrt{S_{n-1} + 1} > 1 + \sqrt{S_{n-1}}$ 则 $\sqrt{S_n} - \sqrt{S_{n-1}} > 1 (n \geq 2)$ 则 $\sqrt{S_n} - \sqrt{S_1} > n-1$ 则 $\sqrt{S_n} > n+1$

则 $S_n > (n+1)^2 (n \geq 2)$ 综上 $S_n \geq (n+1)^2 (n \in \mathbb{N}^*)$

法二: 由 $n=1$ 时, $2\sqrt{S_1} = a_1$ 则 $a_1 = 4$ 或 0 (舍) 则 $S_1 = 4 \geq (1+1)^2$ 成立

假设 $n=k (k \in \mathbb{N}^*)$, $S_k \geq (k+1)^2$ 成立. 由 $2\sqrt{S_{k+1}} = a_{k+1}$ 则 $4S_{k+1} = a_{k+1}^2$ 则 $4S_k + 4a_{k+1} = a_{k+1}^2$

则 $a_{k+1}^2 - 4a_{k+1} + 4 = 4(S_k + 1)$ 则 $(a_{k+1} - 2)^2 = 4(S_k + 1)$ 则 $a_{k+1} - 2 = 2\sqrt{S_k + 1}$

则 $a_{k+1} = 2\sqrt{S_k + 1} + 2 \geq 2\sqrt{(k+1)^2 + 1} + 2 > 2\sqrt{(k+1)^2} + 2 = 2(k+1) + 2 = 2k+4$ 则 $a_{k+1} > 2k+4$

则 $S_{k+1} = S_k + a_{k+1} > (k+1)^2 + 2k+4 = (k+2)^2 + 1 > (k+2)^2$ 则 $S_{k+1} > (k+2)^2$

则 $n=k$ 成立时, $n=k+1$ 也成立. 综上 $S_n \geq (n+1)^2$

(3) 由 $2\sqrt{S_n} = a_n + c$ 则 $S_n = (\frac{a_n + c}{2})^2$ 则 $a_{n+1} = S_{n+1} - S_n = (\frac{a_{n+1} + c}{2})^2 - (\frac{a_n + c}{2})^2$

又由 $a_{n+1} = a_n q$ 则 $(q^2 - 1)a_n^2 + 2a_n \cdot c(q-1) = 4qa_n$ 由 $a_n > 0$

则 $(q^2 - 1)a_n + 2c(q-1) = 4q$ 当 $q=1$ 时, 则 $4=0$ 不成立

当 $q \neq \pm 1$ 时, 则 $-4c = -4$ 则 $c=1$ 当 $q \neq \pm 1$ 时, $a_n = \frac{4q - 2c(q-1)}{q^2 - 1}$ 则 $\{a_n\}$ 为常数列

这与 $q \neq 1$ 矛盾. 则 $c=1$ 为所求

例15: (1) 由 $a_{n+1} - a_n = a_n^2 - 2a_n + 1 = (a_n - 1)^2 > 0$ 则 $\{a_n\}$ 单调递增, 则 $\{\frac{1}{a_n}\}$ 单调递减.

$$(2) \text{ 由 } a_{n+1} - 1 = a_n^2 - a_n = a_n(a_n - 1) \text{ 则 } \frac{1}{a_{n+1} - 1} = \frac{1}{a_n(a_n - 1)} = \frac{1}{a_n - 1} - \frac{1}{a_n}$$

$$\text{则 } \frac{1}{a_n} = \frac{1}{a_n - 1} - \frac{1}{a_{n+1} - 1} \text{ 则 } S_n = \frac{1}{a_1 - 1} - \frac{1}{a_{n+1} - 1} = 1 - \frac{1}{a_{n+1} - 1} \text{ 由 } a_1 = 2, \text{ 则 } a_2 = a_1^2 - a_1 + 1$$

$$= 3 \text{ 由 } \{a_n\} \text{ 单调递增 则 } a_{n+1} \geq a_2 = 3 \text{ 则 } S_n = 1 - \frac{1}{a_{n+1} - 1} \geq 1 - \frac{1}{3-1} = \frac{1}{2}$$

$$\text{由 } a_{n+1} - 1 = a_n(a_n - 1) \text{ 则 } \frac{a_n - 1}{a_{n+1} - 1} = \frac{1}{a_n} \text{ 则 } T_n = \frac{a_1 - 1}{a_{n+1} - 1} = \frac{1}{a_{n+1} - 1} \leq \frac{1}{3-1} = \frac{1}{2}$$

$$(3) \text{ 由 } \frac{1}{T_n} - S_n = (a_{n+1} - 1) - (1 - \frac{1}{a_{n+1} - 1}) = (a_{n+1} - 1) + \frac{1}{a_{n+1} - 1} - 1 > 2\sqrt{(a_{n+1} - 1) \cdot \frac{1}{(a_{n+1} - 1)}} - 1$$

$$= 1 \text{ 则 } \frac{1}{T_n} - S_n > 1$$

例16: (1) 由 $a_{n+1} - a_n = -a_n^2 < 0$, 则 $\{a_n\}$ 单调递减. 又由 $\frac{1}{a_{n+1}} - \frac{1}{a_n} = \frac{1}{a_n - a_n^2} - \frac{1}{a_n} = \frac{1}{a_n(1-a_n)} - \frac{1}{a_n} = \frac{1-(1-a_n)}{a_n(1-a_n)} = \frac{1}{1-a_n}$ 由 $\{a_n\}$ 单调递减, 则 $a_n \leq a_1 = \frac{1}{2}$ 则 $\frac{1}{a_{n+1}} - \frac{1}{a_n} = \frac{1}{1-a_n} \leq \frac{1}{1-\frac{1}{2}} = 2$

$$\text{即 } \frac{1}{a_{n+1}} - \frac{1}{a_n} \leq 2 \text{ 由 } a_{n+1} = a_n(1-a_n) \text{ 又由 } a_n \leq \frac{1}{2} \text{ 则 } 1-a_n > 0 \text{ 即 } a_{n+1} \text{ 与 } a_n \text{ 同号}$$

$$\text{又由 } a_1 = \frac{1}{2} \text{ 则 } a_n > 0 \text{ 则 } \frac{1}{a_{n+1}} - \frac{1}{a_n} = \frac{1}{1-a_n} > \frac{1}{1-0} = 1 \text{ 即 } \frac{1}{a_{n+1}} - \frac{1}{a_n} > 1 \text{ 综上: } 1 < \frac{1}{a_{n+1}} - \frac{1}{a_n} \leq 2$$

$$(2) \text{ 由 } a_{n+1} = a_n - a_n^2 \text{ 则 } a_n^2 = a_n - a_{n+1} \text{ 则 } a_1^2 + \dots + a_n^2 = a_1 - a_2 + a_2 - a_3 + \dots + a_n - a_{n+1} = a_1 - a_{n+1}$$

$$= \frac{1}{2} - a_{n+1} \text{ 又由 } \frac{1}{a_{n+1}} - \frac{1}{a_n} \leq 2 \text{ 则 } \frac{1}{a_{n+1}} - \frac{1}{a_1} \leq 2 \times n \text{ 又由 } a_1 = \frac{1}{2} \text{ 则 } \frac{1}{a_{n+1}} \leq 2n+2 \text{ 则 } a_{n+1} \geq \frac{1}{2n+2}$$

$$\text{则 } \sum_{i=1}^n a_i^2 = \frac{1}{2} - a_{n+1} \leq \frac{1}{2} - \frac{1}{2n+2} = \frac{1}{2} (1 - \frac{1}{n+1}) = \frac{n}{2(n+1)} = \frac{n}{2n+2}$$

例17: (1) 由 $a_1 = m = 17$, 则 $a_2 = 3a_1 + 1 = 52$, $a_3 = \frac{a_2}{2} = 26$, $a_4 = \frac{a_3}{2} = 13$, $a_5 = 3a_4 + 1 = 40$,

$$a_6 = \frac{a_5}{2} = 20, a_7 = \frac{a_6}{2} = 10, a_8 = \frac{a_7}{2} = 5, a_9 = 3a_8 + 1 = 16, \text{ 则 } a_{10} = \frac{a_9}{2} = 8, a_{11} = \frac{a_{10}}{2} = 4,$$

$$a_{12} = \frac{a_{11}}{2} = 2, a_{13} = \frac{a_{12}}{2} = 1 \text{ 则 } n=13 \text{ 为所求}$$

(2) 由 $a_8 = 1$, 若 a_7 为奇数, 由 $a_8 = 3a_7 + 1$, 则 $a_7 = 0$ (舍); 若 a_7 为偶数, 由 $a_8 = \frac{a_7}{2}$,

则 $a_7 = 2$. 若 a_6 为奇数, 由 $a_7 = 3a_6 + 1$, 则 $a_6 = \frac{1}{3}$ (舍); 若 a_6 为偶数, 由 $a_7 = \frac{a_6}{2}$, 则

$a_6 = 4$. 若 a_5 为奇数, 由 $a_6 = 3a_5 + 1$, 则 $a_5 = 1$; 若 a_5 为偶数, 由 $a_6 = \frac{a_5}{2}$ 则 $a_5 = 8$

若 a_4 为奇数, 由 $a_5 = 3a_4 + 1$, 又由 $a_5 = 1$ 或 8 , 则 $a_4 = 0$ (舍) 或 $\frac{7}{3}$ (舍); 若 a_4 为偶数,

由 $a_5 = \frac{a_4}{2}$, 又由 $a_5 = 1$ 或 8 , 则 $a_4 = 2$ 或 16 . 若 a_3 为奇数, 由 $a_4 = 3a_3 + 1$, 又由 $a_4 = 2$ 或 16 ,

则 $a_3 = \frac{1}{3}$ (舍) 或 5 ; 若 a_3 为偶数, 由 $a_4 = \frac{a_3}{2}$, 又由 $a_4 = 2$ 或 16 , 则 $a_3 = 4$ 或 32 .

若 a_2 为奇数, 由 $a_3 = 3a_2 + 1$, 又由 $a_3 = 4$ 或 5 或 32 , 则 $a_2 = 1$ 或 $\frac{4}{3}$ (舍) 或 $\frac{31}{3}$ (舍);

若 a_2 为偶数, 由 $a_3 = \frac{a_2}{2}$, 又由 $a_3 = 4$ 或 5 或 32 , 则 $a_2 = 8$ 或 10 或 64

若 a_1 为奇数, 由 $a_2 = 3a_1 + 1$, 又由 $a_2 = 1$ 或 8 或 10 或 64 , 则 $a_1 = 0$ (舍) 或 $\frac{7}{3}$ (舍) 或 3 或 21

若 a_1 为偶数, 由 $a_2 = \frac{a_1}{2}$, 又由 $a_2 = 1$ 或 8 或 10 或 64 , 则 $a_1 = 2$ 或 16 或 20 或 128

综上 $a_1 \in \{2, 3, 16, 20, 21, 128\}$

$$a_{n+1} = f(a_n) \quad \text{非线性, 变系数, 非齐次}$$

第一步:

非线性 $\rightarrow$ 线性

$$\downarrow$$

$$a_{n+1} = p(n) a_n + Q(n) \quad \text{线性, 变系数, 非齐次}$$

$$\downarrow$$

$$\frac{a_{n+1}}{\prod_{k=1}^n p(k)} = \frac{a_n}{\prod_{k=1}^{n-1} p(k)} + \frac{Q(n)}{\prod_{k=1}^n p(k)}$$

第二步:

变系数 $\rightarrow$ 常系数

$$\text{设 } \frac{a_n}{\prod_{k=1}^{n-1} p(k)} = b_n, \quad \frac{Q(n)}{\prod_{k=1}^n p(k)} = R(n) - R(n+1)$$

$$b_{n+1} = b_n + R(n) - R(n+1) \quad \text{线性, 常系数, 非齐次}$$

$$\downarrow$$

$$b_{n+1} + R(n+1) = b_n + R(n)$$

第三步:

非齐次 $\rightarrow$ 齐次

$$\text{设 } b_n + R(n) = C_n$$

$$\downarrow$$

$$C_{n+1} = C_n \quad \text{线性, 常系数, 齐次}$$

$$\text{由 } \frac{Q(n)}{\prod_{k=1}^n p(k)} = R(n) - R(n+1),$$

$$\text{则 } \left\{ \begin{array}{l} \frac{Q(n-1)}{\prod_{k=1}^{n-1} p(k)} = R(n-1) - R(n) \\ \dots \\ \frac{Q(2)}{\prod_{k=1}^2 p(k)} = R(2) - R(3) \\ \frac{Q(1)}{\prod_{k=1}^1 p(k)} = R(1) - R(2) \end{array} \right. \leftarrow \text{则 } \sum_{j=1}^{n-1} \frac{Q(j)}{\prod_{k=1}^j p(k)} = R(1) - R(n)$$

$$\text{由 } \cancel{b_{n+1} = b_n + R(n)} \quad \text{由 } C_{n+1} = C_n \quad \text{则 } C_n = C_1 \quad \text{则 } b_n + R(n) = b_1 + R(1)$$

$$\text{则 } b_n = b_1 + R(1) - R(n) = b_1 + \sum_{j=1}^{n-1} \frac{Q(j)}{\prod_{k=1}^j p(k)}$$

$$\text{又由 } b_1 = a_1, \quad b_n = \frac{a_n}{\prod_{k=1}^{n-1} p(k)}$$

$$\text{则 } \frac{a_n}{\prod_{k=1}^{n-1} p(k)} = a_1 + \sum_{j=1}^{n-1} \frac{Q(j)}{\prod_{k=1}^j p(k)}$$

$$\text{即 } a_n = \prod_{k=1}^{n-1} p(k) \left(a_1 + \sum_{j=1}^{n-1} \frac{Q(j)}{\prod_{k=1}^j p(k)} \right)$$

(14) 由 $a_n = (-1)^n (4n-3)^2$, 设 $b_n = 4n-3$, 当 n 为偶数, 则 $S_n = -b_1 + b_2 - b_3 + b_4 + \dots + (-1)^{n-1} b_{n-1} + a_n$

$$= (b_2 - b_1)(b_2 + b_1) + (b_4 - b_3)(b_4 + b_3) + \dots + (b_n - b_{n-1})(b_n + b_{n-1}) = 4 \times (b_1 \text{ 缺个平方}) = 4 \times \frac{(1+4n-3)n}{2} = 4n(2n-1)$$

当 n 为奇数, 则 $S_n = S_{n+1} - a_{n+1} = 4(n+1)[2(n+1)-1] - [4(n+1)-3]^2 = 8(n+1)^2 - 8(n+1) + 3$

$$= 8n^2 + 8n + 8 - 8n - 8 + 3 = 8n^2 + 3$$

4

(17) 由 $\cos(\frac{n\pi}{4})$ 周期 $T = \frac{2\pi}{\frac{\pi}{4}} = 8$, 则把每四项看作一个整体 由 $a_n = n \cos(\frac{n\pi}{4}) + 1$
 设 $n = 4k, n = 4k-1, n = 4k-2, n = 4k-3$ 其中 $k \in \mathbb{N}^*$

则 $a_{4k} = 4k \cos(\frac{4k\pi}{4}) + 1 = 4k + 1$; $a_{4k-1} = (4k-1) \cos(\frac{(4k-1)\pi}{4}) + 1 = 4k \cos(\frac{(4k-1)\pi}{4}) + 1 = 1$;
 $a_{4k-2} = (4k-2) \cos(\frac{(4k-2)\pi}{4}) + 1 = (4k-2) \cos(\frac{(4k-2)\pi}{4}) + 1 = -4k + 3$; $a_{4k-3} = (4k-3) \cos(\frac{(4k-3)\pi}{4}) + 1 = 4k \cos(\frac{(4k-3)\pi}{4}) + 1 = 1$

当 $n = 4k$, 由 $a_{4k-3} + a_{4k-2} + a_{4k-1} + a_{4k} = 1 + (-4k + 3) + 1 + 4k + 1 = 6$

则 $S_n = S_{4k} = (a_1 + a_2 + a_3 + a_4) + \dots + (a_{4k-3} + a_{4k-2} + a_{4k-1} + a_{4k}) = 6k$

当 $n = 4k-1$, $S_n = S_{4k} - a_{4k} = 6k - (4k + 1) = 2k - 1$

当 $n = 4k-2$, 由 $S_n = S_{4k-1} - a_{4k-1} = 2k - 1 - 1 = 2k - 2$

当 $n = 4k-3$, 由 $S_n = S_{4k-2} - a_{4k-2} = 2k - 2 - (-4k + 3) = 6k - 5$

综上 $S_n = \begin{cases} 6k-5, & n=4k-3 \\ 2k-2, & n=4k-2 \\ 2k-1, & n=4k-1 \\ 6k, & n=4k \end{cases}$ 其中 $k \in \mathbb{N}^*$

(18) 由 $a_n = |23 - 7n|$ 设 $b_n = 23 - 7n$, 令 $b_n \geq 0$, 则 $n \leq \frac{23}{7}$ 又由 $n \in \mathbb{N}^*$, 则 $n \leq 3$

当 $n \leq 3$ 且 $n \in \mathbb{N}^*$ 时, $b_n \geq 0$; 当 $n > 3$ 且 $n \in \mathbb{N}^*$ 时, $b_n < 0$

当 $n \leq 3$ 且 $n \in \mathbb{N}^*$ 时, $S_n = b_1 + \dots + b_n = \frac{(16 + 23 - 7n) \times n}{2} = \frac{(39 - 7n)n}{2}$; 当 $n > 3$ 时, $S_n = b_1 + b_2 + b_3 - (b_4 + b_5 + \dots + b_n) = - (b_4 + b_5 + \dots + b_n) + 2(b_1 + b_2 + b_3) = -\frac{(39 - 7n)n}{2} + 2 \times \frac{(39 - 7 \times 3) \times 3}{2} = 54 + \frac{(7n - 39)n}{2}$

当 n 为偶数时, $S_n = (a_1 + a_2 + \dots + a_{n-1}) + (a_n + a_{n+1} + \dots + a_n) = \frac{(n-1 + 2n-1) \times \frac{n}{2}}{2}$

当 n 为奇数时, $S_n = S_{n-1} - a_{n-1} = \frac{1}{2}(n-1) + \frac{1}{2}(2^{n-1} - 1) - 2^{n-1} = \frac{1}{2}(n-1) - 2^{n-1}$

当 n 为偶数时, $S_n = \frac{1}{2}(n-1) + \frac{1}{2}(2^{n-1} - 1) - 2^{n-1} = \frac{1}{2}(n-1) - 2^{n-1}$

当 n 为奇数时, $S_n = \frac{1}{2}(n-1) + \frac{1}{2}(2^{n-1} - 1) - 2^{n-1} = \frac{1}{2}(n-1) - 2^{n-1}$

综上 $S_n = \begin{cases} \frac{1}{2}(n-1) + \frac{1}{2}(2^{n-1} - 1) - 2^{n-1}, & n \text{ 为偶数} \\ \frac{1}{2}(n-1) - 2^{n-1}, & n \text{ 为奇数} \end{cases}$

例3: (1) 由 $4S_n = (a_n + 1)^2$ ① 则 $4S_{n+1} = (a_{n+1} + 1)^2$ ② 由 ② - ① 则 $4a_{n+1} = (a_{n+1} + 1)^2 - (a_n + 1)^2$

则 $(a_n + 1)^2 = (a_{n+1} + 1)^2 - 4a_{n+1} = (a_{n+1} - 1)^2$ 则 $a_n + 1 = a_{n+1} - 1$ 或 $a_n + 1 = 1 - a_{n+1}$

若 $a_n + 1 = 1 - a_{n+1}$ 则 $a_{n+1} = -a_n$ 与 $\{a_n\}$ 为正项数列不符, 舍

若 $a_n + 1 = a_{n+1} - 1$ 则 $a_{n+1} - a_n = 2$ 又由 $4S_1 = (a_1 + 1)^2$ 则 $(a_1 - 1)^2 = 0$ 则 $a_1 = 1$

则 $a_n = 1 + 2(n-1) = 2n - 1$

(2) 由 $4S_n = (a_n + 1)^2$ ①, 则 $4S_{n+1} = (a_{n+1} + 1)^2$ ② 由 ② - ① 得 $4a_{n+1} = (a_{n+1} + 1)^2 - (a_n + 1)^2$

则 $(a_n + 1)^2 = (a_{n+1} + 1)^2 - 4a_{n+1} = (a_{n+1} - 1)^2$ 则 $a_n + 1 = a_{n+1} - 1$ 或 $a_n + 1 = 1 - a_{n+1}$

若 $a_n + 1 = 1 - a_{n+1}$ 则 $a_{n+1} = -a_n$ 这与 $\{a_n\}$ 为正项数列不符, 舍

若 $a_n + 1 = a_{n+1} - 1$ 则 $a_{n+1} - a_n = 2$ 当 $n=1$ 时, $4S_1 = (a_1 + 1)^2$ 则 $a_1 = 3$ 则 $a_2 - a_1 = 2$

则 $a_{n+1} - a_n = 2$ 又由 $a_1 = 3$ 则 $a_n = 2n + 1$

4k-1 应改为 4k-3

3,

- 例 2: (1) 由 $a_n = \frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$, 则 $S_n = a_1 + a_2 + \dots + a_n = \frac{1}{1} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \dots + \frac{1}{n} - \frac{1}{n+1} = 1 - \frac{1}{n+1} = \frac{n}{n+1}$
- (2) 由 $a_n = \frac{1}{n(n+2)} = \frac{1}{2}(\frac{1}{n} - \frac{1}{n+2})$, 则 $S_n = \frac{1}{2}(\frac{1}{1} - \frac{1}{3} + \frac{1}{2} - \frac{1}{4} + \dots + \frac{1}{n} - \frac{1}{n+2}) = \frac{1}{2}(\frac{1}{1} + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2}) = \frac{1}{2}(\frac{3}{2} - \frac{1}{n+1} - \frac{1}{n+2})$
- (3) 由 $a_n = \frac{1}{n^2-1} = \frac{1}{(n-1)(n+1)} = \frac{1}{2}(\frac{1}{n-1} - \frac{1}{n+1})$, 则 $S_n = \frac{1}{2}(\frac{1}{1} - \frac{1}{3} + \frac{1}{2} - \frac{1}{4} + \dots + \frac{1}{n-1} - \frac{1}{n+1}) = \frac{1}{2}(1 - \frac{1}{n+1}) = \frac{n}{2n+2}$
- (4) 由 $a_n = \frac{n^2}{4n^2-1} = \frac{1}{4} + \frac{1}{4(n-1)} - \frac{1}{4(n+1)}$, 则 $S_n = \frac{n}{4} + \frac{1}{4}(\frac{1}{1} - \frac{1}{3} + \frac{1}{2} - \frac{1}{4} + \dots + \frac{1}{n-1} - \frac{1}{n+1}) = \frac{n}{4} + \frac{1}{4}(\frac{1}{2} - \frac{1}{n+1}) = \frac{n}{4} + \frac{1}{8} - \frac{1}{4(n+1)}$
- (5) 由 $a_n = \frac{1}{\sqrt{n+1} + \sqrt{n}} = \sqrt{n+1} - \sqrt{n}$, 则 $S_n = (\sqrt{2} - \sqrt{1}) + (\sqrt{3} - \sqrt{2}) + \dots + (\sqrt{n+1} - \sqrt{n}) = \sqrt{n+1} - 1$
- (6) 由 $a_n = \frac{1}{\sqrt{n+1} + \sqrt{n}} = \frac{1}{\sqrt{n+1} + \sqrt{n}} \cdot \frac{\sqrt{n+1} - \sqrt{n}}{\sqrt{n+1} - \sqrt{n}} = \frac{\sqrt{n+1} - \sqrt{n}}{n+1-n} = \frac{\sqrt{n+1} - \sqrt{n}}{1} = \sqrt{n+1} - \sqrt{n}$, 则 $S_n = (\sqrt{1} - \sqrt{0}) + (\sqrt{2} - \sqrt{1}) + \dots + (\sqrt{n+1} - \sqrt{n}) = \sqrt{n+1}$
- (7) 由 $a_n = \frac{1}{\sqrt{n+1} + \sqrt{n}} = \frac{1}{\sqrt{n+1} + \sqrt{n}} \cdot \frac{\sqrt{n+1} - \sqrt{n}}{\sqrt{n+1} - \sqrt{n}} = \frac{\sqrt{n+1} - \sqrt{n}}{n+1-n} = \frac{\sqrt{n+1} - \sqrt{n}}{1} = \sqrt{n+1} - \sqrt{n}$, 则 $S_n = (\sqrt{1} - \sqrt{0}) + (\sqrt{2} - \sqrt{1}) + \dots + (\sqrt{n+1} - \sqrt{n}) = \sqrt{n+1}$
- (8) 由 $a_n = \frac{1}{n(n+1)(n+2)} = \frac{1}{2}(\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)})$, 则 $S_n = \frac{1}{2}(\frac{1}{1 \cdot 2} - \frac{1}{2 \cdot 3} + \frac{1}{2 \cdot 3} - \frac{1}{3 \cdot 4} + \dots + \frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}) = \frac{1}{2}(\frac{1}{1 \cdot 2} - \frac{1}{(n+1)(n+2)}) = \frac{1}{2}(\frac{1}{2} - \frac{1}{(n+1)(n+2)}) = \frac{1}{4} - \frac{1}{2(n+1)(n+2)}$
- (9) 由 $a_n = \frac{2 \times 3^n}{(3^n+1)(3^{n+1}+1)} = \frac{1}{3^n+1} - \frac{1}{3^{n+1}+1}$, 则 $S_n = (\frac{1}{3^0+1} - \frac{1}{3^1+1}) + (\frac{1}{3^1+1} - \frac{1}{3^2+1}) + \dots + (\frac{1}{3^{n-1}+1} - \frac{1}{3^n+1}) = \frac{1}{3^0+1} - \frac{1}{3^n+1} = \frac{1}{4} - \frac{1}{3^n+1}$
- (10) 由 $a_n = \frac{n+2}{n(n+1) \cdot 2^n} = \frac{1}{n} - \frac{1}{n+1} + \frac{1}{2^n}$, 则 $S_n = (\frac{1}{1} - \frac{1}{2} + \frac{1}{2^1}) + (\frac{1}{2} - \frac{1}{3} + \frac{1}{2^2}) + \dots + (\frac{1}{n} - \frac{1}{n+1} + \frac{1}{2^n}) = 1 - \frac{1}{n+1} + \frac{1}{2} - \frac{1}{2^{n+1}} = \frac{1}{2} - \frac{1}{n+1} + \frac{1}{2^{n+1}}$
- (11) 由 $a_n = \frac{n-2}{(n+1)(n+2)} = \frac{1}{n+1} - \frac{1}{n+2}$, 则 $S_n = (\frac{1}{2} - \frac{1}{3}) + (\frac{1}{3} - \frac{1}{4}) + \dots + (\frac{1}{n+1} - \frac{1}{n+2}) = \frac{1}{2} - \frac{1}{n+2}$
- (12) 由 $a_n = \frac{2n+1}{n^2(n+1)^2} = \frac{1}{n^2} - \frac{1}{(n+1)^2}$, 则 $S_n = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{2^2} - \frac{1}{3^2} + \dots + \frac{1}{n^2} - \frac{1}{(n+1)^2} = 1 - \frac{1}{(n+1)^2}$
- (注: $\frac{A}{n^2} - \frac{A}{(n+1)^2} = \frac{A(n+1)^2 - A n^2}{n^2(n+1)^2} = \frac{A(2n+1)}{n^2(n+1)^2}$, 则 $A=1$)
- (13) 由 $a_n = (-1)^n(4n-3)$, 设 $b_n = 4n-3$, 当 n 为偶数, 则 $S_n = -b_1 + b_2 - b_3 + b_4 + \dots + (-1)b_{n-1} + b_n = \frac{1}{2}(b_n - b_{n-1}) = \frac{1}{2} \times 4 = 2n$ 当 n 为奇数, 则 $n+1$ 为偶数, 则 $S_n = S_{n+1} - a_{n+1} = 2(n+1) - (4(n+1)-3) = 1-2n$ 综上 $S_n = \begin{cases} 2n, & n \text{ 为偶数} \\ 1-2n, & n \text{ 为奇数} \end{cases}$
- (14) 由 $a_n = (-1)^n(4n-3)^2$, 设 $b_n = 4n-3$, 当 n 为偶数, 则 $S_n = -b_1^2 + b_2^2 - b_3^2 + b_4^2 + \dots + (-1)b_{n-1}^2 + b_n^2 = (b_2 - b_1)(b_2 + b_1) + (b_4 - b_3)(b_4 + b_3) + \dots + (b_n - b_{n-1})(b_n + b_{n-1}) = 4 \times (b_1 + b_2 + \dots + b_n) = 4 \times \frac{(1+4n-3)n}{2} = 4n(2n-1)$ 当 n 为奇数, 则 $S_n = S_{n+1} - a_{n+1} = 4(n+1)[2(n+1)-1] - [4(n+1)-3]^2 = 8(n+1)^2 - 8(n+1) + 3$
- (15) 由 $a_n = n \cdot 2^n$ 设 $n \cdot 2^n = [A(n+1) + B] \times 2^{n+1} - [A(n+1) + B] \times 2^n = (A+2B) \times 2^n$ 则 $A+B=0$ 则 $A=1$ 则 $S_n = (1 \times 2^1 + 2 \times 2^2 + \dots + n \times 2^n) = (n-1)2^{n+1} + 2$

化简少了n



(17) 由 $q_{n+1} = \frac{a_n - 1}{a_n + 3}$, 设 $a_{n+1} + \lambda = \frac{a_n - 1}{a_n + 3} + \lambda = \frac{(\lambda+1)a_n + 3\lambda - 1}{a_n + 3}$ 则 $\frac{1}{a_{n+1} + \lambda} = \frac{a_n + 3}{(\lambda+1)a_n + 3\lambda - 1} = \frac{\frac{1}{a_n} + \frac{3}{a_n + 1}}{\lambda+1 + \frac{3\lambda-1}{a_n+1}}$
 则 $\lambda = \frac{3\lambda-1}{\lambda+1}$ 则 $\lambda = 1$ 则 $\frac{1}{a_{n+1} + 1} = \frac{\frac{1}{a_n} + \frac{3}{a_n + 1}}{2} = \frac{1}{2a_n + 1} + \frac{1}{2}$ 则 $\frac{1}{a_{n+1}} = \frac{1}{a_n + 1} + \frac{1}{2(n-1)} = \frac{2n-1}{2}$
 则 $a_{n+1} = \frac{2}{2n-1}$ 则 $a_n = \frac{2}{2n-3} - 1 = \frac{7-3n}{2n-3}$

(18) 由 $a_{n+1} = a_n^2$, 则 $19a_{n+1} = 19a_n^2 = 219a_n$ 则 $19a_n = 2^{n-1}19a_1 = (19 \times 2) \times 2^{n-1}$ 则 $a_n = 10^{(19 \times 2) \times 2^{n-1}}$
 $= (10^{19 \times 2})^{2^{n-1}} = 2^{2^{n-1}}$

(19) 由 $a_{n+2} = 5a_{n+1} - 6a_n$, 设 $a_{n+2} + \lambda a_{n+1} = (5+\lambda)(a_{n+1} + \lambda a_n)$ 则 $a_{n+2} = 5a_{n+1} + \lambda(5+\lambda)a_n$
 则 $(5+\lambda)\lambda = -6$ 则 $\lambda = -3$ 或 -2 不妨取 $\lambda = -2$ 则 $a_{n+2} - 2a_{n+1} = 3(a_{n+1} - 2a_n)$
 则 $a_{n+1} - 2a_n = 3^{n-1}(a_2 - 2a_1)$ 则 $a_{n+1} - 2a_n = -3^{n-1}$ 则 $a_{n+1} = 2a_n - 3^{n-1}$ 设 $a_{n+1} + \lambda \cdot 3^{n-1} = 2(a_n + \lambda \cdot 3^{n-2})$
 则 $a_{n+1} = 2a_n - 3\lambda \cdot 3^{n-1}$ 则 $-3\lambda = -1$ 则 $\lambda = \frac{1}{3}$ 则 $a_n + 3^{n-1} = 2^{n-1}(a_1 + 3^{-1}) = 3 \times 2^{n-1}$
 则 $a_n = 3 \times 2^{n-1} - 3^{n-1}$

(20) 由 $a_{n+2} = 4a_{n+1} - 4a_n$, 设 $a_{n+2} + \lambda a_{n+1} = (4+\lambda)(a_{n+1} + \lambda a_n)$ 则 $a_{n+2} = 4a_{n+1} + \lambda(4+\lambda)a_n$
 则 $(4+\lambda)\lambda = -4$ 则 $\lambda = -2$ 则 $a_{n+2} - 2a_{n+1} = 2(a_{n+1} - 2a_n)$ 则 $a_{n+1} - 2a_n = 2^{n-1}(a_2 - 2a_1)$
 则 $a_{n+1} - 2a_n = -2^{n-1}$ 则 $a_{n+1} = 2a_n - 2^{n-1}$ 同除 2^{n+1} 则 $\frac{a_{n+1}}{2^{n+1}} = \frac{a_n}{2^n} - \frac{1}{4}$ 则 $\frac{a_n}{2^n} = \frac{a_1}{2} - \frac{1}{4}(n-1)$
 则 $\frac{a_n}{2^n} = \frac{5-n}{4}$ 则 $a_n = (5-n) \times 2^{n-2}$

(21) 由 $a_{n+1} = \begin{cases} 2a_n, & n \text{ 为奇数} \\ a_{n+1} + 1, & n \text{ 为偶数} \end{cases}$ 当 n 为偶数时, 则 $n+1$ 为奇数, 则 $a_{n+1} = a_n + 1, a_{n+2} = 2a_{n+1}$
 则 $a_{n+2} = 2(a_n + 1) = 2a_n + 2$ 则 $a_{n+2} - 2 = 2(a_n + 2) - 2 = 2(a_n + 1) = 2^{n-1}(a_2 + 2) = 2^{n-1} \times 3 = 2^n$
 则 $a_n = 2^{\frac{n-2}{2}} - 2$ (n 为偶数) 当 n 为奇数时, 则 $n+1$ 为偶数, 则 $a_{n+1} = 2a_n$
 则 $a_n = \frac{1}{2}a_{n+1} = \frac{1}{2}(2^{\frac{n+1}{2}} - 2) = 2^{\frac{n-1}{2}} - 1$ 综上 $a_n = \begin{cases} 2^{\frac{n-1}{2}} - 1, & n \text{ 为奇数} \\ 2^{\frac{n-2}{2}} - 2, & n \text{ 为偶数} \end{cases}$

(22) 当 n 为偶数时, 则 $n+1$ 为奇数, 则 $a_{n+1} + a_n = 2n+1, a_{n+2} - a_{n+1} = 2(n+1)$
 则 $a_{n+2} = a_{n+1} + 2(n+1) = (2n - a_n) + 2(n+1) = -a_n + 4n + 2$ 设 $a_{n+2} + \lambda(n+2) + \mu = -(a_n + \lambda(n+1) + \mu)$
 则 $a_{n+2} = -a_n - 2\lambda n - 2\mu - 2\lambda$ 则 $\begin{cases} -2\lambda = 4 \\ -2\mu - 2\lambda = 2 \end{cases}$ 则 $\begin{cases} \lambda = -2 \\ \mu = 1 \end{cases}$ 则 $a_n - 2n + 1 = (-1)^{\frac{n}{2}-1}(a_2 - 2 \times 2 + 1)$
 由 $a_2 - a_1 = 2 \times 1$, 则 $a_2 = 4$ 则 $a_n = (-1)^{\frac{n}{2}-1} + 2n - 1$ (n 为偶数)

当 n 为奇数时, 则 $n+1$ 为偶数, 则 $a_{n+1} - a_n = 2n$ 则 $a_{n+1} = a_n + 2n = [(-1)^{\frac{n-1}{2}-1} + 2(n-1) - 1] + 2n = (-1)^{\frac{n-1}{2}} + 1$ (n 为奇数)
 综上 $a_n = \begin{cases} (-1)^{\frac{n-1}{2}} + 1, & n \text{ 为奇数} \\ (-1)^{\frac{n}{2}-1} + 2n - 1, & n \text{ 为偶数} \end{cases}$

(23) 当 n 为偶数时, 则 $n+1$ 为奇数, 则 $a_{n+1}a_n = 4n^2 - 1, a_{n+2}a_{n+1} = 4(n+1)^2 - 1$, 则 $\frac{a_{n+2}}{a_n} = \frac{2n+3}{2n-1} = \frac{2(n+2)-1}{2n-1}$
 则 $\frac{a_n}{a_2} = \frac{2n-1}{2 \times 2 - 1} = \frac{2n-1}{3}$, 又由 $a_1a_2 = 4 \times 1^2 - 1 = 3$, 则 $a_2 = \frac{3}{2}$ 则 $a_n = \frac{2n-1}{2} = n - \frac{1}{2}$ (n 为偶数)
 当 n 为奇数时, 则 $n+1$ 为偶数, 则 $a_na_{n+1} = 4n^2 - 1$, 则 $a_n = \frac{4n^2-1}{a_{n+1}} = \frac{(4n^2-1)}{4(n+1)^2-1} = 2(2n-1) = 4n-2$ (n 为奇数)
 则 $a_n = \begin{cases} n - \frac{1}{2}, & n \text{ 为偶数} \\ 4n-2, & n \text{ 为奇数} \end{cases}$

首项算错



例1: (1) 设 $\{a_n\}$ 的第 p 项 ($p \in \mathbb{N}^*$) 与 $\{b_n\}$ 的第 q 项 ($q \in \mathbb{N}^*$) 相等, 则 $a_p = b_q$
 则 $3p-1 = 2q+1$ 则 $q = \frac{3p-2}{2}$ 由 $q \in \mathbb{N}^*$, 则 $3p-2$ 为偶数, 则 p 为偶数 则 $C_n = a_{2n} = 3 \times (2n) - 1 = 6n - 1$

(2) 设 $\{a_n\}$ 的第 p 项 ($p \in \mathbb{N}^*$) 与 $\{b_n\}$ 的第 q 项 ($q \in \mathbb{N}^*$) 相等, 则 $a_p = b_q$ 则 $3p-1 = 2^q$
 则 $p = \frac{2^q+1}{3} = \frac{(3-1)^q+1}{3} = \frac{1}{3} [\sum_{i=0}^q C_q^i 3^i (-1)^{q-i} + (-1)^q + 1] = \sum_{i=0}^q C_q^i 3^{i-1} (-1)^{q-i} + \frac{(-1)^q+1}{3}$ 由 $p \in \mathbb{N}^*$, 则 $\frac{(-1)^q+1}{3}$ 应为整数 则 q 为奇数 则 $C_n = b_{n-1} = 2^{n-1}$

(3) 设 $\{a_n\}$ 的第 p 项 ($p \in \mathbb{N}^*$) 与 $\{b_n\}$ 的第 q 项 ($q \in \mathbb{N}^*$) 相等, 则 $a_p = b_q$, 则 $3p-1 = 2^q-1$
 则 $p = 2^{q-1}$ 则 $\forall q \in \mathbb{N}^*$, 都有 $\exists p \in \mathbb{N}^*$, 使 $p = 2^{q-1}$ 则 $\{b_n\}$ 的所有项都在 $\{a_n\}$ 中.
 设 S_{20} 中去掉了 m ($m \in \mathbb{N}^*$) 项 $\{a_n\}$ 与 $\{b_n\}$ 的公共项 则 $S_{20} = a_1 + a_2 + \dots + a_{20} - (b_1 + b_2 + \dots + b_m)$ 由 $b_m \leq a_{30+2m} < b_{m+1}$ 则 $2^m - 1 \leq 2(20+m) - 1 < 2^{m+1} - 1$ 则 $2^m \leq \frac{2m+40}{m+39} < 2^{m+1}$
 则 $m = 5$. 则 $S_{20} = a_1 + \dots + a_{25} - (b_1 + b_2 + \dots + b_5) = \frac{(1+25) \times 25}{2} - [\frac{2(2^5-1)}{2-1} - 5] = 568$

(4) 设 $\{c_n\}$ 中在 a_{k+1} 之前 (不含 a_{k+1}) 所有项的项的个数记为 P_k
 则 $P_k = 2+3+\dots+k+1 = \frac{k(2+k+1)}{2} = \frac{k(k+3)}{2}$ 由 $P_m < 25 < P_{m+1}$ 则 $\frac{m(m+3)}{2} < 25 < \frac{(m+1)(m+4)}{2}$
 则 $m = 5$ 则 $S_{25} = a_1 + a_2 + a_3 + a_4 + a_5 + a_6 + a_7 + a_8 + a_9 + a_{10} + a_{11} + a_{12} + a_{13} + a_{14} + a_{15} + a_{16} + a_{17} + a_{18} + a_{19} + a_{20} + a_{21} + a_{22} + a_{23} + a_{24} + a_{25} - (a_1 + a_2 + \dots + a_5) + (a_1 + 2a_2 + \dots + 5a_5) + 4C_5 = \frac{(1+25) \times 25}{2} + [2+2 \times 2 + \dots + 5 \times 2^5 - (1+2+\dots+5)] + 4 \times (2^5-1) = 531$

(5) 设 $\{c_n\}$ 中 p 项是 $\{a_n\}$ 中的项, q 项是 $\{b_n\}$ 中的项 ($p, q \in \mathbb{N}^*$) 则 $p+q=27$
 若 $C_{27} = 2^q$, 则 $C_{26} = 2^q - 1$ 则 $S_{27} = (2^1 + 2^2 + \dots + 2^q) + (1+3+\dots+2^q-1)$ 由 $\{a_n\}$ 的第 p 项为 2^q-1 则 $2p+1 = 2^q-1$ 则 $p = \frac{2^q-2}{2} = 2^{q-1}-1$ 又由 $p+q=27$ 则 $2^{q-1}-q-28=0$.
 又由 $q \in \mathbb{N}^*$, 则 q 不存在

若 $C_{27} = 2p+1$, 则 $S_{27} = (1+3+\dots+2p+1) + 2^1 + \dots + 2^q$ 其中 $2^q < 2p+1 < 2^{q+1}$, 又由 $p+q=27$
 则 $2^q < 55-2q < 2^{q+1}$ 又由 $q \in \mathbb{N}^*$, 则 $q=5$ 则 $S_{27} = \frac{(1+27) \times 27}{2} + \frac{2(2^5-1)}{2-1} = p(p+1) + 2^{q+1} - 2 = 22 \times 23 + 2^6 - 2 = 568$

(6) 由 $2^k \leq 3n-1 \leq 2^{k+1}$ 则 $\frac{2^k-1}{3} \leq n \leq \frac{2^{k+1}+1}{3}$
 当 k 为奇数时, $\frac{2^k+1}{3}$ 为整数, 则 $2^{k+1}+1$ 除以 3 的余数为 2, 则 $\frac{2^{k+1}+1}{3}$ 整数部分为 $\frac{2^{k+1}+1-2}{3} = \frac{2^{k+1}-1}{3}$ 由 $\frac{2^k+1}{3} \leq n \leq \frac{2^{k+1}+1}{3}$ 则 $C_k = \frac{2^{k+1}-1}{3} - \frac{2^k+1}{3} + 1 = \frac{2^k+1}{3}$ (k 为奇数)
 当 k 为偶数时, $\frac{2^{k+1}+1}{3}$ 为整数, 由 2^{k+1} 除以 3 的余数为 2, 则大于 $\frac{2^k+1}{3}$ 的最小整数为 $\frac{2^k+1}{3} = \frac{2^k+2}{3}$ 则 $C_k = \frac{2^{k+1}+1}{3} - \frac{2^k+2}{3} + 1 = \frac{2^k+2}{3}$ (k 为偶数)
 综上 $C_k = \begin{cases} \frac{2^k+1}{3}, & k \text{ 为奇数} \\ \frac{2^k+2}{3}, & k \text{ 为偶数} \end{cases}$

(7) 由 $b_k = \frac{a_{k+1} - a_k}{k+1} = \frac{2^{k+1} - 2^k}{k+1} = \frac{2^k}{k+1}$

数列字母写错

